Title and Product: Celsite® Implantofix Implant	**Number**	BER 011
	Revision	Ed06

Revision	Date	Description
01	30/04/2020	Creation
02	30/03/2021	Add of CMR / ED Status, MSDS (composition table) Add of Biological risk analysis Update with Biological Tests results Modif A706: deletion of two references Deletion of irrelevant data about spanner Update of PMS data (12-2020)
03	24/08/2022	Modif A792: change of Pellethane 2363-80AE and 2363-90A manufacturing site (Section 3.3) Modif X505: Liveo Brand name Add of biological test results
04	24/11/2023	Layout update (integration of data from annexes in main report) Including Modification n°A912, A953, A957Bis, A976
05	30/04/2024	Modif A1007: Screw connector green colorant production shut down
06	04/06/2024	Modif A1005: Deletion of references

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INTRODUCTION: PURPOSE AND SCOPE

The purpose of this report is to evaluate the overall biological safety of the implantable components of the **Celsite® Venous Access Port Systems** with a PSU access port (i.e. the access port made of PSU (including the silicone septum), the catheter and the screw connector) of the following references:

TABLE 1: CELSITE® IMPLANTOFIX IMPLANT REFERENCES[a]

Component	Reference	Designation
Access port	I3130	CELSITE IMPLANTOFIX S GM ASSEMBLE
	I3131	CELSITE IMPLANTOFIX S PM ASSEMBLE
Connection ring	I2025	VIS IMPLANTOFIX VERTE
	I2355	VIS CELSITE IMPLANTOFIX S
Catheter	A3524	CATHETER PUR 5F / 37 / LZ
	A3601	CATHETER SI M FINI

The **Celsite® Implantofix Implant** are one of the implantable components of the **Celsite® Venous Access Port Systems** in which they are associated with posing accessories. Each **Celsite® Implantofix Implant** contains one access port, one catheter and one screw connector.

The different possible configurations of **Celsite® Implantofix Implant** are presented in the table below:

TABLE 2: CELSITE® IMPLANTOFIX IMPLANT CONFIGURATIONS[A]

Configuration	Implant composition		
	Acces Port	Catheter	Screw connector
ST PUR 5F IV	I3130	A3524	I2025
SM PUR 5F IV	I3131	A3524	I2025
SM SIL 6F IV	I3131	A3601	I2355

This document is presented in 2 parts :

- A Biological Evaluation Plan which includes the description of the device, its raw materials and manufacturing process, the categorization of the device and the potential biological risks associated, the physical and chemical characterization of the device and the biological test program.
- A Biological Evaluation Report which includes the reports of all biological tests conducted and the overall analysis of the results which allows to conclude on the toxicological risk associated to the device

It should be noted that:

- This biological evaluation report (BER) focuses only on **Celsite® Implantofix implant** described in this report, and does not address the biological risks associated with other devices or accessories.
- This BER is not intended to evaluate the sterility of the device or of its packaging. The validation of the sterilization process and compliance with ISO 10993-7 *"Biological Evaluation of Medical devices – Part 7: Ethylene oxide sterilization residuals"* is documented in the kits associated Sterilization Evaluation Report. However, the influence of sterilization and packaging on the biocompatibility is assessed within the risk analysis of the manufacturing process.

[a] See Modification A1005 in Section 7.2

PART 1 : BIOLOGICAL EVALUATION PLAN

1. Standards and references applied for the medical device related to biological evaluation

The list of the applicable documents is presented in Table 3.

TABLE 3 : APPLICABLE DOCUMENTS

Reference	Title
EU MDD	Applicable sections of the European Medical Device Directive 93/42/EEC amended 2007/47/EC
EU MDR	Applicable sections of the European Medical Device Regulation 2017/745/EU (published in Official Journal of the European Union on May 5th, 2017) (GSPR 1, 6, 10.1, 10.2, 10.4)
GMED Technical Memo	Biological evaluation report Information to be provided for assessment
GMED Guide	Biological assessment of medical devices according to the ISO 10993-1 standard
ISO 10993-1 (2018)	Biological evaluation of medical devices – Part 1: Evaluation and testing within a risk management process
ISO 10993-2 (2022)	Biological Evaluation of Medical devices – Part 2: Animal welfare requirements
ISO 10993-3 (2014)	Biological Evaluation of Medical devices – Part 3: Tests for genotoxicity, carcinogenicity and reproductive toxicity
ISO 10993-4 (2017)	Biological evaluation of medical devices — Part 4: Selection of tests for interactions with blood
ISO 10993-5 (2009)	Biological Evaluation of Medical devices – Part 5: Tests for *in vitro* cytotoxicity
ISO 10993-6 (2016)	Biological Evaluation of Medical devices – Part 6: Tests for local effects after implantation
NF EN ISO 10993-7 (2008) /A1 (2022)	Biological Evaluation of Medical devices – Part 7: Ethylene oxide sterilization residuals
ISO 10993-9 (2019)	Biological Evaluation of Medical devices – Part 9: Framework for identification and quantification of potential degradation products
ISO 10993-10 (2020)	Biological Evaluation of Medical devices – Part 10: Tests for skin sensitization
ISO 10993-11 (2017)	Biological Evaluation of Medical devices – Part 11: Tests for systemic toxicity
ISO 10993-12 (2021)	Biological Evaluation of Medical devices – Part 12: Sample preparation and reference materials
ISO 10993-13 (2021)	Biological Evaluation of Medical devices – Part 13: Identification and quantification of degradation products from polymeric medical devices
ISO 10993-15 (2019)	Biological Evaluation of Medical devices – Part 15: Identification and quantification of degradation products from metals and alloys
ISO 10993-16 (2017)	Biological Evaluation of Medical devices – Part 16: Toxicokinetic study design for degradation products and leachables
ISO 10993-17 (2002)	Biological Evaluation of Medical Devices – Part 17: Methods for the establishment of allowable limits for leachable substances
ISO 10993-18 (2020)	Biological Evaluation of Medical devices – Part 18: Chemical characterization of materials
ISO/TS 10993-19 (2020)	Biological evaluation of medical devices – Part 19: Physico-chemical, morphological and topographical characterization of materials
ISO/TS 10993-20 (2006)	Biological evaluation of medical devices – Part 20: Principles and methods for immunotoxicology testing of medical devices
ISO 10993-23 (2021)	Biological evaluation of medical devices – Part 23: Tests for irritation
ASTM F756-17 (2017)	Standard practice for assessment of hemolytic properties of materials
ISO 14971 (2019)	Medical Devices – Application of risk management to medical devices, Annex I: Guidance on Risk Analysis Procedures for Biological Hazards
ISO/IEC 17025 (2017)	General requirements for the competence of testing and calibration laboratories

2. Formulation, description, manufacturing and use of the medical device

The Celsite® IMPLANTOFIX is an implantable device for vascular access intended to be used in adults and children. The Celsite® IMPLANTOFIX allows safe repeated access to patient's blood stream for administration of medications and fluids. The system consists of an access port, with a silicone septum, which is connected to a catheter with a screw connector. The catheter can be made of silicone or polyurethane (PU) depending on the reference. The access to the system is via percutaneous injection using a special

non-coring needle. At the end of treatment and after medical approval, the port and catheter system should be removed. The Celsite® IMPLANTOFIX devices are designed to allow safe and repetitive punctures of the septum for up to five years when used following the conditions described in the instruction for use (IFU) document.

The Celsite® IMPLANTOFIX is available in a range of sizes and is sold in kits. Depending on the kit reference, other accessories not evaluated in this document are optionally added.



FIGURE 1 : CELSITE® IMPLANTOFIX IMPLANT

2.1 Formulation

The description of the raw materials used to manufacture the **Celsite® Implantofix Implant** are provided in the following sections.

The **Celsite® Implantofix Implant** are designed to have an even surface and to be without sharp angles. The tips of the catheter are trimmed in order to be smoothened and rounded.

2.1.1 Access ports

TABLE 4: DESCRIPTION OF RAW MATERIALS OF ACCESS PORTS

Component	Component Manufacturer	Raw Material	Amount (%)	Raw Material Supplier	Type of Patient Contact
Body (I1240, I1241)	Devillé	EVIVA® EV-26000	100	SOLVAY Specialty Polymers S.p.A.	Implant Tissue Long-term (>30 days)
Ring (I12105, I1211)	Devillé	EVIVA® EV-26000	100	SOLVAY Specialty Polymers S.p.A.	Implant Tissue Long-term (>30 days)
Septum (I2300 – I2305)	B. Braun Medical (Chasseneuil)	Liveo™ Q7-4750 Biomedical Grade ETR Elastomer	100	Dow Corning Europe S.A.	Implant Blood path, Indirect Tissue Long-term (>30 days)
ORX ring (I1220, I1221)	Société Parisienne de Ressort (France)	Cobalt-Chromium alloy UGI® PHYNOX® (ISO 5832-7)	100	Ugitech	No contact
Exit cannula (I1252)	Möller Medical GmbH (Allemagne)	Stainless Steel (ISO 5832-1)	100	Schoeller Werk GmbH & Co KG	Implant Blood path, Indirect Long-term (>30 days)

2.1.2 Screw connector

TABLE 5: DESCRIPTION OF RAW MATERIALS OF SCREW CONNECTORS

Component	Component Manufacturer	Raw Material	Amount (%)	Raw Material Supplier	Type of Patient Contact
Outer part (I1025)	Top Clean Injection	EVIVA EV-26000	100	Solvay Specialty Polymers USA	Implant Tissue Long-term (>30 days)
Inner part (6514081)	B. Braun Medical (Chasseneuil)	Pellethane 2363-80AE	100	The Lubrizol Corporation	Implant Tissue Long-term (>30 days)

2.1.3 Catheters

TABLE 6: DESCRIPTION OF RAW MATERIALS OF SILICONE CATHETER

Component		Component Manufacturer	Raw Material	Amount (%)	Raw Material Supplier	Type of Patient Contact
Silicone Catheter M (A3601)	Tube Si (A1602)	DDP Speciality Electronic Materials	Liveo™ RX65R Medical Grade Tubing Radiopaque	100	DDP Specialty Products Germany GmbH & Co. KG	Implant Circulating blood Tissue Long-term (>30 days)
	Ink (A1060)	B. Braun Chasseneuil	RTV MED-6608-2	Trace	Nusil Technology LLC	Implant Circulating blood Tissue Long-term (>30 days)
	Luer connector (A1240)	Sté Escoffier Freres	TECHNYL SAFE A 216FC NC	100	RHODIA Opérations	Ext. Com. Device, Blood path, indirect Limited (<24h)

TABLE 7: DESCRIPTION OF RAW MATERIALS OF POLYURETHANE CATHETER[A]

Component		Component Manufacturer	Raw Material	Amount (%)	Raw Material Supplier	Type of Patient Contact
PUR Catheter 5F/37 LZ (A3524)	Tubing (A1590)	MS Techniques	Tecothane TT2095A-B20	100	The Lubrizol Corporation	Implant Circulating blood Tissue Long-term (>30 days)
	Luer connector (A1240)	Sté Escoffier Freres	TECHNYL SAFE A 216FC NC	100	DOMO Engineering Plastics Europe S.P.A.	Ext. Com. Device, Blood path, indirect Limited (≤24h)
	Tubing connection (A1602)	Freudenberg Medical Europe GmbH	Liveo™ RX65R Medical Grade Tubing Radiopaque	100	DDP Specialty Products Germany GmbH & Co. KG	No contact

2.2 Primary packaging

The **Celsite® Implantofix Implant** is packaged with the accessories of the kit in a rigid thermoformed tray made of polyethylene terephthalate (PET) and a Tyvek lid which are then placed in a breather bag. A batch number label is placed on the breather bag.

The **Celsite® Implantofix Implant** is only in contact with the PET tray and the Tyvek lid when in the package.

TABLE 8: PACKAGING RAW MATERIALS

Component	Material	Supplier
Rigid Thermoformed Tray	PET LAIZE 333 mm	Pentaplast S.A.
Tyvek lid	Tyvek® 1073B Co 702887C	SudPack Medica
Breather Bag	Multilayered film 36603-S and Tyvek 2 FS/CR27	Nelipak Healthcare Packaging Ltd

2.3 Description of the expected and intended biological effect, if applicable

The raw materials used are not intended to have biological effect when implanted in the human body.

No user contact is anticipated as users are supposed to wear gloves during surgery.

2.4 Manufacturing of the medical device

Initial manufacturing steps (assembly of the access port, laser marking of the lot number on the access port body, manufacture of the catheter and molding of the screw connector), as well as primary packaging of the **Celsite® Implantofix Implant** are performed by B. Braun Medical (Production site: Chasseneuil du Poitou, France).

The packed **Celsite® Implantofix Implant** is sterilized using ethylene oxide (EO) by Sterigenics SAS (Rantigny).

The sterilized device is finally sent back to B. Braun Medical (Production site: Chasseneuil du Poitou) where the secondary packaging of the **Celsite® Implantofix Implant** is performed.

The following processing agents are used during the **Celsite® Implantofix Implant** manufacturing process.

TABLE 9: PROCESS AGENTS USED FOR CELSITE® IMPLANTOFIX IMPLANT COMPONENTS MANUFACTURING

Manufacturing Step	Processing Agent Name
Cleaning of the body	Neodisher Mediclean Forte
Cleaning of the ring/ORX ring - Alternative 1	Neodisher Mediclean Forte
Cleaning of the ring/ORX ring - Alternative 2	TopKlean MC20D
Cleaning of the exit cannula - Alternative 1	Neodisher Mediclean Forte
Cleaning of the exit cannula - Alternative 2	TopKlean MC20D
Cleaning of the catheter	Ethanol (70°)
Deburring of the septum	Ethanol (70°)

2.5 Use of the medical device

Target Population: The **Celsite® Implantofix Implants** are intended to be used in adults, children and neonates.

Claimed clinical performance: The **Celsite® Implantofix Implant** allows safe repeated access to patient's blood stream for administration of medications and fluids.

Life duration: The **Celsite® Implantofix Implant** is a single use device. The **Celsite® Implantofix Implant** should be explanted when it's clinically indicated or following standard clinical practices (we recommend 5 years after implantation) taking into account all recommendations listed in the Instructions for Use.

Shelf life: The shelf life of our sterile finished **Celsite® Implantofix Access Port System** including the **Celsite® Implantofix Implant** is 5 years.

Storage condition: The storage condition of the sterile finished **Celsite® Implantofix Access Port System** including the **Celsite® Implantofix Implant** are: "Storage temperature should be between 10°C < T < 35°C. Avoid freezing and excessive heat".

3. Categorization of the medical device: nature and duration of contact

As part of the evaluation process based on the requirements of ISO 10993-1 and 2020 FDA Biocompatibility Guidance, medical devices are categorized based on the type and the duration of patient contact.

When used as intended, the **Celsite® Implantofix Implant** are categorized as follow:
- Category: Implant medical device
- Nature of the contact with the human body: Circulating Blood / Tissue
- Duration of the contact with the human body: Long term (> 30 days)

4. Identification of potential biological risks of the medical device / possible biological hazards

Because of the **Celsite® Implantofix Implant** categorization, consideration must be given at least to all relevant biological endpoints as described in Table 10.

TABLE 10 : ENDPOINTS TO BE CONSIDERED ON THE CELSITE® IMPLANTOFIX IMPLANT

Endpoint	Applicable ISO 10993 Standard	Applicable (Y/N)
Physical and/or Chemical Information	ISO 10993-1	Y
Cytotoxicity	ISO 10993-5	Y
Sensitization	ISO 10993-10	Y
Irritation or Intracutaneous Reactivity	ISO 10993-23	Y
Material Mediated Pyrogenicity	ISO 10993-11	Y
Acute Systemic Toxicity	ISO 10993-11	Y
Subacute Toxicity	ISO 10993-11	Y
Subchronic Toxicity	ISO 10993-11	Y
Chronic Toxicity	ISO 10993-11	Y
Implantation Effects	ISO 10993-6	Y
Hemocompatibility	ISO 10993-4	Y
Genotoxicity	ISO 10993-3	Y
Carcinogenicity	ISO 10993-3	Y
Reproductive/ Developmental toxicity	ISO 10993-3	Y
Immunotoxicity	ISO 10993-20	Y
Degradation	ISO 10993-9, ISO 10993-13, ISO 10993-15	Y

(Y) Endpoints to be considered and documented *per* the ISO 10993-1 standard and the 2020 FDA Biocompatibility Guidance
(N) Not applicable

It is important to note that while all of the endpoints listed above must be considered, they can be addressed in a number of different ways including chemical testing, biological testing, and written justification or risk assessment, if relevant information is available.

5. Physical and chemical information for biological risk analysis / medical device characterization

5.1 Material characterization protocol

Materials without any body contact do not have to be considered according to ISO 10993-1 and therefore they are not part of the material characterization.

Evaluation of the chemical nature of the material can take the form of experimental data and/or information on the chemistry of the materials/components involved. Literature studies conducted on the materials help evaluate the biological response and are useful in assessing a finished medical device for its intended use/intended purpose. Some factors that affect the biocompatibility of the material include the identity, concentration, availability, and toxicity of all constituents such as additives, processing aids, and monomers.

Devices must be designed and manufactured in such a way as to reduce to a minimum the risks posed by substances leaching from the product. Special attention shall be given to substances having Carcinogenic, Mutagenic or Reprotoxic (CMR substances) or endocrine-disrupting properties according to European Regulations[b, c].

Identification of the materials and processes used in the manufacture of a medical device enables the intrinsic toxicity of the device to be investigated. The biological risks arising from the use of medical devices are related to two main sources, regardless of the type and duration of body contact:

- Raw materials used to fabricate the medical device, which can combine different chemicals capable of leaching from the device and/or producing degradation products that cause adverse effects once in contact with tissues and body fluids. Moreover, the surface state, porosity, geometry, aspect (solid, gel, liquid) may also influence the biological reaction in the immediate vicinity of the medical device (tissues, bone, blood), i.e., its local tolerance, hemocompatibility;
- Manufacturing process, including machining, assembly, cleaning, packaging and sterilization, are all steps that can provide residues which may be bioavailable once in contact with the body and body fluids and can be responsible for additional biological hazards.

However, in order to estimate the risk of this occurring, it is necessary to obtain information demonstrating to what extent these constituents are available during intended use. The amounts can be estimated from extraction tests on the material.

A chemical characterization to evaluate any risks will request to determine dose-based threshold for these specific chemicals, with demonstrated ability from the laboratory in charge of the analysis to detect these specific substances at, or below, the chemical-specific Analytical Evaluation Threshold.

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[b] CMR substances according to Table 3.1 of Annex VI of Regulation (EC) 1272/2008 (Version 2022-12-17, consolidated text)

[c] ED Substances according to current ED assessment list and SVHC list from ECHA (European Chemicals Agency): https://www.echa.europa.eu/ed-assessment ; https://www.echa.europa.eu/candidate-list-table

The ISO 10993 series is intended for use by *professionals, appropriately qualified by training and experience, who are able to interpret its requirements and judge the outcome of the evaluation for each medical device, taking into consideration all the factors relevant to the medical device, its intended use and the current knowledge of the medical device provided by review of the scientific literature and previous clinical experience.*

5.2 Characterization of the materials

5.2.1 Risk analysis of Patient-contacting Raw Materials

As presented in Section 2.1, the raw materials in contact (direct or indirect) with the patient are the following:

Raw Material	Raw Material Supplier (Name and Address)	Specifications: MSDS and/or TDS	CAS N°	CMR? ED? SVHC?
EVIVA® EV-26000	**SOLVAY Specialty Polymers S.p.A.** Viale Lombardia, 20 20021, Bollate Italia	3_DK[1] 3_DK(TDS)[2] 210101_FR24_DL_EVIVA[3]	25154-01-2 13463-67-7 1314-13-2	Not listed
Liveo™ Q7-4750 Biomedical Grade ETR Elastomer	**Dow Corning Europe S.A.** Rue Jules Bordet - Parc Industrial - Zone C B-7180 Seneffe Belgium	3_OS[4] 3_OS(TDS)[5] 220602_FR24_COMP_X1450[6] 201030_FR24_PRI_X1450[7]	68083-19-2 7631-86-9 70131-67-8 68083-18-1 67923-19-7 556-67-2 541-02-6 540-97-6 2627-95-4 68478-92-2 68037-59-2 78-27-3	556-67-2 : SVHC (PBT, vPvB), CMR R2 (H361f) 541-02-6 : SVHC (PBT, vPvB) 540-97-6: SVHC (PBT, vPvB)
Stainless Steel (ISO 5832-1)	**Schoeller Werk GmbH & Co KG** Im Kirschseiffen 53940 Hellenthal GERMANY	3_FD[8]	Proprietary	Not listed
Pellethane 2363-80AE	**The Lubrizol Corporation** 29400 Lakeland Boulevard Wickliffe, Ohio 44092 USA	3_AG[9]	Proprietary	Not listed
Liveo™ RX65R Medical Grade Tubing Radiopaque	**DDP Speciality Electronic Materials** 1635 N. Gleaner Road Hemlock, MI 48626 United States	3_IG[10] 3_IG(TDS)[11] 210919_FR24_PRI_A1601[12] 220602_FR24_COMP_A1601[13]	7727-43-7	Not listed
RTV MED-6608-2	**Nusil Technology LLC** 1050 Cindy Lane Carpinteria, California 93013 USA	3_DW[14]	1330-20-7 22984-54-9 77-58-7	Not listed Not listed R1B[d]
TECHNYL SAFE A 216FC NC	**DOMO Engineering Plastics Europe S.P.A.** Via Linfano, 18 38062 ARCO (TN) Italia	3_EZ[15] 3_EZ(TDS)[16]	32131-17-2	Not listed

[d] The amount of the substance represents < 0,1% w/w of the catheter tubing

Raw Material	Raw Material Supplier (Name and Address)	Specifications: MSDS and/or TDS	CAS N°	CMR? ED? SVHC?
Tecothane TT2095A-B20	**The Lubrizol Corporation** 29400 Lakeland Boulevard Wickliffe, Ohio 44092 USA	3_IB[17] 3_DL_3_EL_3_IB(TDS)[18]	Proprietary 7727-43-7 13463-67-7	Not listed

Information provided by the raw material supplier and general information issued from the litterature review are presented in the sections below.

5.2.1.1 Data from suppliers

EVIVA® EV-26000

According to the Technical Data Sheet (3_DK(TDS)[2]), Eviva® EV-26000 is a grade of polysulfone offered exclusively for use in implantable portacaths (central venous access devices). It meets the requirements of ISO 10993-5 ad ISO 10993-18.

According to the SDS provided (3_DK[1]), Eviva® EV-26000 is not classified as hazardous according to the Regulation (EC) No 1272/2008. It contains the following compounds:

Compound	CAS No.	Concentration (%)	Health Hazard Statement Code
Polysulfone	25154-01-2	≥90	N/A
Titanium Dioxide	13463-67-7	≥5 - ≤10	N/A
Zinc Oxide	1314-13-2	≥0.1 - ≤0.25	N/A

According to the Declaration Letter (210101_FR24_DL_EVIVA[3]) provided, Eviva® EV-26000:

- Meets the criteria outlined in the EU MDR 2017/745 regulation (Paragraph 10.4.1) for CMR and endocrine disrupting substances. Substances as outlined in Section 10.4.1 a & b are not contained at concentrations above 0.1% (w/w).
- Does not contain animal fats, oils, milk products or other animal products.
- Contains intentionally added BPA, 4,4'-isopropylidenediphenol (CAS-No. 80-05-7).

Liveo™ Q7-4750 Biomedical Grade ETR Elastomer

According to the Technical Data Sheet (3_OS(TDS)[5]), Liveo™ BioMedical Grade ETR Elastomers (Q7-series) are a series of two-part, enhanced-tearresistant (ETR) silicone elastomers that consist of dimethyl and methylvinyl siloxane copolymers and reinforcing silica. The elastomers are supplied as a twocomponent kit (Part A and B), equal portions (by weight) of which must be thoroughly blended together prior to use. The elastomer is then thermally cured via addition-cure (platinum cure) chemistry. The Liveo™ BioMedical Grade ETR Elastomers (Q7-series) are compatible with Ethylene Oxyde sterilisation.

According to the Declaration letter from the supplier (2206_FR24_COMP_X1450[6]), Liveo™ Q7-4750 Biomedical Grade ETR Elastomer parts A and B main ingredients (before mixed) are as follows:

Part	Chemical	CAS No.	Weight percent
Part A	Dimethyl siloxane, dimethylvinyl-terminated	68083-19-2	50 – 70

Title and Product: Celsite® Implantofix Implant	**Number**	BER 011
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Part	Chemical	CAS No.	Weight percent
	Silicon dioxide	7631-86-9	20 – 30
	Dimethyl siloxane, hydroxyl-terminated	70131-67-8	3 – 7
	Siloxanes and Silicones, di-Me, Me vinyl, vinyl group-terminated	68083-18-1	3 – 7
	Dimethyl, methylvinyl siloxane, hydroxyterminated	67923-19-7	< 0.5
	Octamethyl Cyclotetrasiloxane D4	556-67-2	< 0.1
	Decamethylcyclopentasiloxane D5	541-02-6	< 0.2
	Dodecamethyl cyclohexasiloxane D6	540-97-6	< 0.2
	Tetramethyldivinyldisiloxane	2627-95-4	< 0.1
	Platinum, 1,3-diethenyl-1,1,3,3-tetramethyldisiloxane complexes	68478-92-2	<0.1
Part B	Dimethyl siloxane, dimethylvinyl-terminated	68083-19-2	50 – 70
	Silicon dioxide	7631-86-9	20 – 30
	Dimethyl siloxane, hydroxyl-terminated	70131-67-8	3 – 7
	Siloxanes and Silicones, di-Me, Me vinyl, vinyl group-terminated	68083-18-1	3 – 7
	Dimethyl, methylhydrogen siloxane	68037-59-2	1 – 3
	Dimethyl, methylvinyl siloxane, hydroxyterminated	67923-19-7	< 0.5
	Octamethyl Cyclotetrasiloxane D4	556-67-2	< 0.1
	Decamethylcyclopentasiloxane D5	541-02-6	< 0.2
	Dodecamethyl cyclohexasiloxane D6	540-97-6	< 0.2
	1-Ethynylcyclohexanol	78-27-3	< 0.1

According to the provided Safety Data Sheet (3_OS[4]), Liveo™ Q7-4750 part A and part B silicon elastomers are not considered hazardous according to the European Regulation (EC) No. 1272/2008.

Decamethylcyclopentasiloxane (CAS 541-02-6) and Dodecamethylcyclohexasiloxane (CAS 540-97-6) are classified as Substances of Very High Concern (SVHC) and are present at levels above 0,1%.

Dodecamethylcyclohexasiloxane (D6) and decamethylcyclopentasiloxane (D5) have been identified by the Member State Committee of ECHA as fulfilling the vPvB (very Persistent and very Bio-accumulative) criteria laid down in Annex XIII to Regulation (EC) No 1907/2006.

The estimated LD_{50} for oral route is > 5000 mg/kg and for dermal route is > 2000 mg/kg (estimation, no animal species mentioned). Silicone Liveo™ Q7-4750 part A and part B are not irritating to the skin but can be an irritant to the eyes.

According to the Product Regulatory Information (201030_FR24_PRI_X1450[7]), Liveo™ Q7-4750 Biomedical Grade ETR Elastomer does not contain:

- Allergens, including cereals, crustaceans, eggs, fish, peanuts, soybeans, milk, other nuts, celery, mustard, sesame, sulphur dioxide and sulphites, lupins, and molluscs;
- Animal Derived Materials,
- Antibiotics;
- Cytotoxins;

- GMO as defined in 1829/2003/EC and 1830/2003/EC;
- Genotoxic Impurities;
- Hormones;
- Residual Solvents;
- Metal wastes (lead, cadmium, mercury, and hexavalent chromium);
- Waste Electrical & Electronic Equipment (substances listed in Annex VII of Directive 2012/19/EU);
- Substances prohibited in accordance with Directive 2011/65/EU and Directive 2015/863/EU, nor lead, mercury, hexavalent chromium, PBB, PBDE (including decaBDE, PentaBDE or OctaBDE), Phtalates (including DEHP, BBP, DBP and DIBP) above the specified concentrations of 0.1% (by weight) and cadmium above 0.01% (by weight).
- Nanotechnology;
- Any chemicals listed by the State of California under the Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65) as being known to cause cancer, birth defects or other reproductive harm;
- Added antioxydants;
- Added pigments;
- Added preservatives;
- Aflatoxin;
- Abestos;
- Bisphenol A (CAS No. 80-05-7);
- Butylated Hydroxyanisole (CAS No. 25013-16-5);
- Butylated Hydroxytoluene (CAS No. 128-37-0);
- Dioxin;
- Ethylene Oxide;
- Gluten;
- Glycerin or other Jatropha derived materials;
- Lactose;
- Latex;
- Melamine;
- 2-Mercaptobenzothiazole (CAS No. 149-30-4);
- Palm oil;
- Phthalates;
- Polychlorinated Biphenyls;
- Polybrominated Byphenyls;
- Carcinogenic, mutagenic or toxic to reproduction (‘CMR’) substances, of category 1A or 1B, or substances having endocrine-disrupting properties as defined in paragraph 10.4.1 of MDR (EU) 2017/745;
- Nitrosamines.

It is worth noting that component A and B never come into contact with the patient in this way. As component A and B react with each other, the individual ingredients of components A and B are no longer present in the final product. Therefore, chemical and biological tests are conducted on the final product to asses its safety.

Liveo™ RX 65R Medical Grade Tubing Radiopaque

Acording to the Product Regulatory Information provided (210919_FR24_PRI_A160), Liveo™ RX65R Medical Grade Tubing Radiopaque Special is white silicone tubing made from tear-resistant Liveo™ brand

Title and Product: Celsite® Implantofix Implant	**Number**	BER 011
	Revision	Ed06

BioMedical Grade platinum-cured elastomer containing barium sulfate. It is designed for use in medical devices anywhere an extruded silicone profile is needed. It does not contain:

- Allergens, including cereals, crustaceans, eggs, fish, peanuts, soybeans, milk, other nuts, celery, mustard, sesame, sulphur dioxide and sulphites, lupins, and molluscs;
- Animal Derived Materials,
- Antibiotics;
- Cytotoxins;
- GMO as defined in 1829/2003/EC and 1830/2003/EC;
- Genotoxic Impurities;
- Hormones;
- Residual Solvents;
- Metal wastes (lead, cadmium, mercury, and hexavalent chromium);
- Waste Electrical & Electronic Equipment (substances listed in Annex VII of Directive 2012/19/EU);
- Substances prohibited in accordance with Directive 2011/65/EU and Directive 2015/863/EU, nor lead, mercury, hexavalent chromium, PBB, PBDE (including decaBDE, PentaBDE or OctaBDE), Phtalates (including DEHP, BBP, DBP and DIBP) above the specified concentrations of 0.1% (by weight) and cadmium above 0.01% (by weight).
- Nanotechnology;
- Any chemicals listed by the State of California under the Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65) as being known to cause cancer, birth defects or other reproductive harm;
- Added antioxydants;
- Added pigments;
- Added preservatives;
- Aflatoxin;
- Abestos;
- Bisphenol A (CAS No. 80-05-7);
- Butylated Hydroxyanisole (CAS No. 25013-16-5);
- Butylated Hydroxytoluene (CAS No. 128-37-0);
- Dioxin;
- Ethylene Oxide;
- Gluten;
- Glycerin or other Jatropha derived materials;
- Lactose;
- Latex;
- Melamine;
- 2-Mercaptobenzothiazole (CAS No. 149-30-4);
- Palm oil;
- Phthalates;
- Polychlorinated Biphenyls;
- Polybrominated Byphenyls;
- Carcinogenic, mutagenic or toxic to reproduction (‘CMR’) substances, of category 1A or 1B, or substances having endocrine-disrupting properties as defined in paragraph 10.4.1 of MDR (EU) 2017/745;
- Nitrosamines.

According to the Declaration Letter on the composition of Liveo™ RX65R Medical Grade Tubing Radiopaque (220602_FR24_COMP_A1601), Studies were conducted on tubing of different durometers to assess the

concentrations of extractable D4, D5 and D6 under conditions that reasonably reflect the intended use. Tubing specimens were extracted in 75% Ethanol/25% Water For Injection (WFI) solvent. Extractions took place at 45°C for 7 days. D4, D5 and D6 levels were not above reporting ECHA SVHC reporting threshold of 0.1%.

According to the Technical Data Sheet (3_IG(TDS)), Liveo™ RX65R Medical Grade Tubing Radiopaque is made from BioMedical Grade elastomer tht exceeds USP Class VI Plastics Test Requirements and it meets ISO 10993 guidelines for body contact applications.

According to the Safety Data Sheet (3_IG), the product has not been classified as hazardous according to Regulation (EC) No 1272/2008.

RTV MED-6608-2

According to the Safety Data Sheet provided (3_DW[14]), MED-6608-2 is a silicone dispersion mixture composed of the following substances:

- 15 - 25% Reaction mass of ethylbenzene and xylene (No CAS N°)
- 5 - 15% 2-Butanone, O,O',O''-(methylsilylidyne)trioxime (CAS N°22984-54-9)
- < 0.3% Dibutyltin dilaurate (CAS N°77-58-7)

It is Harmful in contact with skin. Harmful if inhaled. Causes skin irritation. Causes serious eye irritation. May cause an allergic skin reaction. May cause respiratory irritation. May cause damage to organs through prolonged or repeated exposure.

It contains no substance on the REACH candidate list.

TECHNYL SAFE A 216FC NC

According to the technical Data Sheet provided (3_EZ(TDS)[16]), TECHNYL SAFE A 216FC NC is an unreinforced polyamide 66, with standard viscosity, for injection moulding. It offers all of the primary properties of unreinforced polyamide 66: thermal and mechanical properties, chemical resistance, impact and abrasion resistance. It is designed to be used in food contact applications.

According to the Safety Data Sheet (3_EZ[15]), TECHNYL SAFE A 216FC NC is not classified as a hazardous substance or mixture according to Regulation EC No 1272/2008. It is composed of polyamide 6.6 (CAS N°32131-17-2). It does not contain susbtances classified as SVHC at concentration ≥ 0,1%.

Polyamide 6.6 is not considered hazardous for acute toxicity, not irritant for skin or eyes, it is not a skin sensitizer. It is not genotoxic or carcinogenic, nor toxic for reproduction.

Tecothane TT2095A-B20

According to information provided by The Lubrizol Corporation, Tecothane® TT2095A-B20 is not classified as hazardous according to Regulation EC No. 1272/2008.

According to information provided by The Lubrizol Corporation, Tecothane® TT2095A-B20 has no known health hazards.

It contains 10-20% of Barium Sulfate (CAS N°7727-43-7) and 0,1-1% of Titanium Dioxide (CAS N°13463-67-7).

The following toxicological data is provided in the MSDS (3_IB[17]):

Acute toxicity:

- Oral : May cause irritation of the gastrointestinal tract. Not classified for acute toxicity based on available data.
- Dermal: Not classified for acute toxicity based on available data.
- Inhalation: Overexposure to vapors or mist may cause dizziness, headache, nausea, and/or flu-like symptoms. Persons with sensitive airways (e.g., asthmatics) may react to vapors. Not classified for acute toxicity based on available data.
- Skin Corrosion/Irritation: Not classified as a primary skin irritant. Prolonged or repeated skin contact as from clothing wet with material may cause dermatitis. Symptoms may include redness, edema, drying, and cracking of the skin.
- Serious Eye Damage/Eye Irritation: Not classified as a primary eye irritant.
- Respiratory sensitization:
    - Product: Under decomposition conditions, isocyanates may be generated from this product. Isocyanates can cause skin sensitization and/or respiratory sensitization.
    - Titanium dioxide: Not a respiratory sensitizer (Literature)
- Skin sensitization:
    - Product: Under decomposition conditions, isocyanates may be generated from this product. Isocyanates can cause skin sensitization and/or respiratory sensitization.
    - Titanium dioxide: Not a skin sensitizer. (Literature)
- Specific Target Organ Toxicity - Single Exposure: No data available
- Aspiration Hazard:
    - Titanium dioxide: Not classified
- Other effects: This product contains titanium dioxide which is bound/encapsulated within the polymer. It is not expected to present a hazard when this product is used, handled, and processed.

Chronic Effects:

- Carcinogenicity:
    - Product: Not available.
    - Titanium dioxide: Suspected of causing cancer.
- Germ Cell Mutagenicity: No data available
- Reproductive toxicity:
    - Titanium dioxide: Not classified
- Specific Target Organ Toxicity - Repeated Exposure:
    - Product: Prolonged inhalation of high levels of barium sulfate dust may cause lung damage resulting in a benign pneumoconiosis.
    - Barium sulfate: Prolonged inhalation of high levels of barium sulfate dust may cause lung damage resulting in a benign pneumoconiosis.

5.2.1.2 Data from litterature

General information issued from the literature are summarised in the following Table 11:

TABLE 11: SUMMARY OF LITERATURE REVIEW

Raw Material	Protocol / Report number	Conclusion of the review
Polysulfone	PRO-03-2022-v01[19] REP-03-2022-v01[20]	PSU resin are widely used in various applications in field of medical devices. Owing to their thermal and chemical stability they are considered stable and biologically inert. Majority of toxicity data available on the polymer is related to the sensitization, cytotoxicity, hemocompatibility and implantation studies. The use of PSU resin as pacemaker or hemofilters produced severe immune reactions in isolated cases of

Raw Material	Protocol / Report number	Conclusion of the review
		allergic reactions to the material. The polymer was found non-cytotoxic in different cell-lines. The hemocompatibility of the PSU dialyzer is been well established by the available literature. As implantable material PSU ophthalmic lenses pose a hazard of corneal opacity and local toxicity owing to nutritional deficiency of the corneal tissue. The data on implantation of PSU composites as orthopedic material revealed limitation of the polymer as prostheses for bone remodeling but not for joint replacement purpose. The polymers are also studied as base material for artificial tissue preparations that can substitute the allografts. It also finds application as the dental implants. Following the process of bibliographic search in all databases using the keywords defined in the protocol, literature on systemic toxicity endpoints like single-dose, repeat-dose, carcinogenicity, reproductive and developmental toxicity, and genotoxicity cannot be retrieved. Although the data on health hazards associated with the use of PSU is not available in the public domain, its widespread use in medical devices symbolizes their reasonable biocompatibility. As the data supporting the biocompatibility of PSU resin is not available per ISO-10993 requirement, the use of the polymer as a class 3 implant will require thorough testing of all the toxicity endpoints.
Titanium Dioxide	PRO-19-2022-v01[21] REP-19-2022-v01[22]	Titanium dioxide (titania, TiO2) is chemically inert material. It is of low acute toxicity by all routes of exposure relevant to human health. It is non-irritating and non-sensitizer in animal studies. Oral and inhalation studies in animals have shown that there are no adverse effects associated with repeated exposure of titanium dioxide. Genotoxicity tests were positive for titanium dioxide however, positive response in genotoxicity studies were considered to be due to cytotoxic effects. Carcinogenicity studies showed no potential for carcinogenicity associated with titanium dioxide. However, considering effects of titanium dioxide in lungs after repeated exposure it is suspected to be a carcinogen via inhalation route. It is not a reproductive toxicant. Based on the studies on cytotoxicity and hemocompatibility related effects are majorly driven by the particle size of the material used. Further, Titanium dioxide is included on the list of FDA-approved Inactive ingredient.
Zinc Oxide	PRO-07-2022-v01[23] REP-07-2022-v01[24]	ZnO has low acute toxicity after oral and inhalation exposure. ZnO is not a skin irritant. Although single and repeated inhalation exposures to ultra-fine ZnO fumes showed changes in pulmonary function and induced airway inflammation (metal fume fever), none of the studies allow the establishment of a NOAEL for metal fume fever with a reasonable degree of certainty. ZnO is not a skin irritant. Data in guinea pigs and humans indicate that ZnO is not a very potent sensitizing agent. This is supported by the fact that Zn compounds, especially ZnO have been used for over decades in a variety of pharmaceutical and cosmetic products (some of them even dermatological preparations against skin irritation) without any such reported effects. Data provided on the repeated oral dose toxicity of ZnO indicate adverse effect on food intake and neuroglia cells. The inhalation related toxic effects are assigned to degenerative changes in nasal cavity and oedematous changes in lungs. Several data were provided on the genotoxicity of ZnO. Data on other Zn compounds have also been used, based on the assumption that after intake the biological activities of the Zn compounds are determined by the Zn cation. The available data indicate that the genotoxicity results vary widely. Conflicting results have been found, even in the same test systems. Overall, the results of the in vitro tests indicate that Zn has genotoxic potential in vitro based on positive results in mammalian test systems for gene mutations and chromosomal aberrations. The positive result for chromosomal aberrations in vitro is considered overruled by negative in vivo tests for this endpoint. The limited data available indicate that Zn deficiency or supplementation may influence carcinogenesis, since promoting and inhibiting actions have been reported. However,

Raw Material	Protocol / Report number	Conclusion of the review
		there is no clear experimental or epidemiological evidence for a direct carcinogenic action of Zn or its compounds. Some data were provided on the reproductive toxicity of ZnO. Male rats suffered reduced sperm motility. However, no such effect was observed on female reproductive organs. Developmental toxicity studies do not permit the derivation of a proper NOAEL because neither reproductive nor developmental or maternal effects were observed. In some animals' foetal resorption and malformation was observed which was not dose dependant. Cytotoxic studies carried out with ZnO showed positive response at higher doses. The ZnO-eugenol mixture seems to have cytotoxic as well as inflammatory response in in vivo implantation studies. Whereas the newly developed Zn alloys appears to be hemocompatible and produce less tissue reaction to support the use of Zn alloys as implantable devices.
Silicone	PRO-06-2022-v01[25] REP-06-2022-v01[26]	Silicone elastomer and silicone fluid is of low acute toxicity by all routes of exposure relevant to human health. In animal studies, components of silicone elastomer are non-irritating. Further, intracutaneous study conducted in rabbits with silicone elastomer extract shows that it is non-irritating. In different studies, components of silicone elastomer were found to be sensitizer. However, based on guinea pig maximisation test provided in unpublished data shows that silicone elastomer is non-sensitizing. Oral studies in animals shows that no adverse effect of silicone derivatives upon oral administration. Inhalation studies in animals shows chronic inhalation exposure to silicone derivatives can cause cancer and reproductive toxicity. However, the mechanisms by which the inhalation exposure triggers the toxicity are not relevant for humans. Therefore, no concern related to human health hazards is expected. Genotoxicity data available on silicone derivatives suggests that they are not mutagenic. Further, bacterial reverse mutation study performed with silicone elastomer showed that it was not mutagenic. Dietary carcinogenicity studies in animals concluded that silicone derivatives are not carcinogenic in either species. Inhalation exposure of animals to silicone derivatives results in carcinogenicity and are classified as carcinogen to humans. Silicone and its derivatives are not toxic to reproduction in animal studies. However, the potential impurities Octamethylcyclotetrasiloxane and Decamethylcyclopentasiloxane are considered reproductive toxicant in animal studies. These effects are not considered relevant for human considering non-relevant mechanisms. Silicone elastomers are not cytotoxic in in vitro study. Further, multiple cytotoxicity studies show that Silicone elastomer or its extractable or leachable are also non-cytotoxic. Hemocompatibility study using different modification of silicone elastomer shows that they are compatible. Similar results are observed with silicone elastomer in unpublished study. Implantation study with silicone elastomer in animal showed some adverse effects. However, 1-week and 12-week implantation study with shows that silicone elastomers are safe for implantation. Pyrogenicity study reported in unpublished study report shows that silicone elastomer is non-pyrogenic. Based on the literature available for silicone elastomer and unpublished study reports available for the materials, no safety concern is expected from the material from a toxicological perspective.
Stainless steel	PRO-11-2022[27] REP-11-2022[28]	The current literature review report on stainless steel concludes that 304 SS and 316L SS possess very low toxicity. Based on GHS/CLP, stainless steels should be classified as carcinogen category 2, and specific target organ toxicity repeated exposure 1 (STOT RE1) according to their nickel content. But we found that the nickel release rate from the 316L SS is below 0.5 µg/cm2/week in artificial sweat.1 Moreover, 28 days 316L SS

Raw Material	Protocol / Report number	Conclusion of the review
		inhalation toxicity study showed no adverse effects.15 Furthermore, chromium(III)-rich oxide formed on the surface of stainless steel prevents the release of metals even in harsh physiological conditions. The literature review and history of stainless steels in medical devices revealed that stainless steels (304 SS and 316L SS) are non-carcinogen. Implanted metallic medical devices can release ions and debris into the body. Studies showed that the release of metallic ions from stainless steel is always within the acceptable limit. Various animal studies evaluate the ability of stainless steel to induce the local tumor at the site of implantation and found no local tumor development at the site. The studies from stainless steel grinding, polishing and manufacturing have not raised any major concerns about the potential carcinogenicity of stainless steel. The constituents present in the stainless steel do not have any reproductive toxicity properties, so the occurrence of reproductive toxicity in animals or humans with stainless steel is much less. To conclude, various toxicological endpoints discussed in the report show that stainless steels (304 SS and 316L SS) are not toxic and highly biocompatible in nature. The literature evaluated also reflects that there are currently no alternative materials with a better combination of physical properties and historical database of use.
Barium Sulfate	PRO-17-2022-v01[29] REP-17-2022-v01[30]	Literature on safety/toxicity of barium sulfate is minimal. However, based on the available literature information, barium sulfate is known to have low or no toxicity because of its insoluble nature. The insoluble, nontoxic nature of barium sulfate has made it practical to use this particular barium compound in medical applications as a contrast media and is frequently used as a benign, radiopaque aid to x-ray diagnosis in colorectal and some upper gastrointestinal examinations, because it is normally not absorbed by the body after oral intake. Under these routine medical situations, barium sulfate is generally safe. However, barium sulfate may potentially be toxic when it is introduced into the gastrointestinal tract under conditions where there is colon cancer or perforations of the gastrointestinal tract and barium is able to enter the blood stream. Based on the available literature search and the data available for toxicity endpoints like acute, repeated dose toxicity, irritation, sensitization, genotoxicity, carcinogenicity, reproductive and development toxicity, for barium sulfate or its structural analogues it can be concluded that barium sulfate has a low toxicity potential with no adverse effects and can be safely used in medical applications. Also, barium sulfate with dermal fibroblast cells showed to be noncytotoxic. Barium sulfate (BaSO4) was the first radiopaque material to be widely compounded in medical formulations and is the most common filler used with medical-grade polymers and has a long history of safe use, thus concluding its safety.
Polyurethane	PRO-21-2022[31] REP-21-2022[32]	Polyurethanes represent a wide spectrum of polymeric materials that are used in several biomedical applications. The composite nature of polyurethanes facilitates the materials to be adjustable in terms of their properties and structure. Polyurethanes, based on their chemical compositions and structure, can be used in a wide range of devices, from permanent implants to temporary biodegradable scaffolds promoting tissue regeneration. The versatility of the polymer is also attributed to its properties such as robustness, flexibility and resistance. The literature review suggests that the polymer class has an overall positive biocompatibility. However, owing to the complexity of implantable polyurethanes and the host of responses possible in the body environment, particular attention must be given to the choice of polyurethanes and their composition while keeping in mind the intended use of the device. Polyurethane foams are also associated with hazardous additives and flame retardants that may be carcinogenic or mutagenic. The degradation products of PURs may be hazardous to the environment and occupational workers. The starting materials of Polyurethane, diisocyanates, are also associated with specific adverse health effects such as respiratory tract irritations and breathing impairment and

Raw Material	Protocol / Report number	Conclusion of the review
		is often associated with asthma and bronchitis. However, exposures to isocyanates are not expected to be of concern and magnitude in cured polymers. Degradation of Polyurethanes may occur when implanted in the body for several years. Such phenomena are of particular significance in presence of metallic particles such as cobalt, which are known to catalyse such degradation. Several approaches like surface treatments and coating, grafting techniques, impregnation processes and additives, crosslinking and photo-chemical immobilization reactions have been evaluated to improve the antithrombotic properties and overall biostability of polyurethanes.

5.2.1.3 Summary of risk analysis of Patient-contacting Raw Materials

The collected data from the suppliers, as well as the data obtained from literature search, allows to identify that some of them are identified as of potential concerns. These concerns are either from the raw materials themselves, or from the substances that could be hazardous if they are released into the patients.

For the list of concerns provided below, the hazards (only) are being identified:

- The raw material EVIVA® EV-26000 contains titanium dioxide ≥ 5 - ≤ 10%, which is considered as carcinogenic in particles form in case of inhalation – this hazard is considered there as non clinically relevant. EVIVA® EV-26000 also contains zinc oxide, which is potential sensitizer and irritant and is associated with conflicting results in genotoxicity evaluations (not mutagenic in Ames test, but positive in in vivo chromosomal aberration and MLA). Polysulfone EVIVA® EV-26000 is made with Bisphenol A associated with skin sensitization potential, and which is also a substance of very high concern due to its endocrine disruptive properties, and which is classified Reprotoxic 1B.
- The Liveo™ Q7-4750 Biomedical Grade ETR Elastomer, Liveo™ Q7-4780 and MED2-4502 may contain some traces of D4 classified Reprotoxic Category 2 by the European Union Regulation (EC) No 1272/2008. It may contain also traces of D5, which is a mild irritant to the skin.
- TECHNYL SAFE A 216FC NC, Barium Sulfate, Unalloyed Titanium are of no major health concerns in regards to its history of use in implants and other medical devices.
- RTV MED-6608-2 is not expected to cause hazards considering ony traces of the substance are in contact with the patient.
- The Tecothane TT2095A-B20 of the catheters is not expected to be of concerns per se however, aromatic polyurethane may degrade and be responsible for release of leachables

5.2.2 Risk analysis of Manufacturing Process

5.2.2.1 Processing aids

CAS numbers of the processing aids are used to determine whether there are any classifications relating to human health according to the European Medical Device Regulation 2017/745/EU or the EC regulation 1272/2008 as amended.

When possible, the processing aids have been reviewed for their Summary of Classification and Labelling (CLP) status, either as reported in the Safety Data Sheets (SDS) or in the European Regulation (EC) No. 1272/2008 (CLP Regulation) as amended.[b] The noted properties of concern were irritation, sensitization, mutagenicity, carcinogenicity, and reproductive toxicity (including lactation effects).

The Endocrine Disrupting (ED) properties of each chemical were also assessed based on information, if any, detailed within the REACH *Registry of SVHC intentions* or *endocrine disruptor assessment*.[c]

The summary of Health Hazard Statement Codes and ED assessment on the processing aids used during the manufacture of **Celsite Implantofix Implant** is summarized in Table 12.

TABLE 12 : PROCESSING AIDS – SUMMARY OF HEALTH HAZARD STATEMENT CODES AND ENDOCRINE DISRUPTION

Processing Aids	Composition	CAS Number	Health Hazard Statement Code*	CMR? ED? SVHC?
ALCOOL ETHYLIQUE 70% (PA0002)[33]	Mixture	N/A	H319	N/A
	Distilled water	7732-18-5	N/A	N/A
	Ethanol	64-17-5	H319	N/A
NEODISHER MEDICLEAN FORTE (PA0021)[34]	Mixture	N/A	N/A	N/A
	alcools gras alcoxylés (≥0.1-<1)	120313-48-6	H315	N/A
SOLVANT A3 TOPKLEAN MC20D (PA0033)[35]	Mixture	N/A	H315, H319	N/A
	3-butoxypropan-2-ol (≥80)	5131-66-8	H315, H319	N/A

(n/a): chemical is not listed in the queried documents.
(*): Health hazard statement code as mentioned in the SDS
Health hazard codes: **H315**: Causes skin irritation ; **H319**: Causes serious eye irritation

Based on information presented in Table 12, no processing aids as prepared and used in the manufacture of the device contain substances classified CMR 1A or 1B or with endocrine-disrupting properties.

Some processing agents are classified as harmful or toxic. Toxicity for eye contact is considered to be non-clinically relevant since the final product is not in contact with the eyes.

Health hazards such as skin irritation are not fully relevant to the clinical use of the device. However, it cannot be excluded that residues of these manufacturing agents may have an adverse effect upon clinical use of the device.

As the **Celsite® Implantofix Implant** are commercialized, the manufacturing process and cleaning is validated to remove or minimize the amount of unwanted residues of manufacturing. Therefore, the patient would unlikely be exposed to these processing aids at level of toxicological concerns.

To assess the biological risks related to potential processing aids residues, any chemical characterization or biological test are conducted on a finished device which has followed all manufacturing process steps.

5.2.2.2 Primary Packaging

The sterility of the **Celsite® Implantofix Implant** is maintained by the packaging until its point-of-use.

Any contaminants present on the surface of the primary packaging may migrate to the medical device, essentially to the outer surface of the medical device itself. Based on this, consideration must also be given to the primary packaging.

According to the packaging raw material supplier information:

- Tyvek® 1073B Co 7502887C is made from 100% HDPE and is suitable for packaging medical device. Tyvek® 1073B Co 7502887C meets the requirements of ISO 10993-5 and is compatible with EO sterilization. All components of Tyvek® 1073B Co 7502887C comply with FDA Regulation.
- PET films are extruded rigid films without plasticizers. PET LAIZE 333 mm comply with the most recent package requirements for heavy metals of the Toxic Packaging Cleaning House (TPCH, formerly CONEG) legislation and the latest March 9, 2015, requirements of Directive 2013/2/EU, as well as the Commission Decision of 2011/534/EU amending Directive 2002/95/EC (RoHS-Reduction of Hazardous substances) in their actual valid version.
- The Breather bag is composed of a multilayered film and a Tyvek® widow
    - The Multilayered film 36603-S is composed of the following layers: LLDPE / PE / Nylon / PE / Nylon / PE / LLDPE
    - Tyvek® 2FS®/CR27 is made from 100% HDPE with CR27 coating and is suitable for packaging medical device and is compatible with EO sterilization.

The primary packaging materials are provided by well-known suppliers of packaging materials for medical devices. It is reasonable to consider that the selected materials are good materials to reduce individual biological risk by design as recommended by EN ISO 14971 principles.

In order to assess the biological risks related to the primary packaging, any chemical characteization or biological test are conducted on a finished device packaged in its primary packaging.

5.2.2.3 Sterilization

The packed **Celsite® Implantofix Implant** is sterilized using ethylene oxide (EO) by Sterigenics SAS (Rantigny).

EO sterilization is known to produce highly reactive residues of ethylene oxide, CAS No. 75-21-8, classified and labeled with CMR health hazard codes H350 (Carc. 1B), H340 (Muta. 1B) and H360Fd (Repr. 1B) *per* the 14th ATP of the EU Regulation (EC) No. 1272/2008.

Compliance with ISO 10993-7, including evaluation of sterilization-related residues present on the final, finished device, is documented in the associated Sterilization Evaluation Report. Ethylene sterilization processes are validated and the ethylene oxide sterilization residuals meet the ISO 10993-7 requirements.

It is considered herein that the biocompatibility of the devices may be affected by the sterilization process; therefore, only testing of a final, finished (*i.e.*, sterile) device can confirm that the sterilization process does not affect the biocompatibility of the device.

Furthermore, the ethylene oxide sterilization process requires heat and humidity that can increase the migration of substances from the primary packaging to the surface of the device.

Therefore, taking into account that the finished device can be re-sterilized if necessary, all chemical and biological tests are conducted on finished devices sterilized twice by ethylene oxide.

5.2.2.4 Summary of risk analysis of Manufacturing process

After an analysis of the hazard identification of the manufacturing processes used to construct the **Celsite® Implantofix Implant**, it appears that no processing aids are associated with classifications of greatest toxicological concern (SVHC, CMR 1A or 1B substances). However, some hazards such as irritation where identified and need to be adressed.

The primary packaging of the **Celsite® Implantofix Implant** is made of PET thermoformed tray and a Tyvek lid place in a breather bag made of polyethylene film and Tyvek. These raw materials are usual in primary packaging of medical devices, they are well characterized with a long history of clinical use in approved and marketed medical devices packaging.

Sterilization by Ethylene Oxide is known to produce highly active residues. The **Celsite® Implantofix Implant** are sterilized with validated ethylene oxide sterilization processes and the ethylene oxide sterilization residuals meet the ISO 10993-7 requirements.

While the manufacturing process is expected to minimize, and eliminate as much as possible, the quantity of unwanted residues, the combined residual risks as defined by ISO 14971 cannot be addressed by a literature review since manufacturing residues are specific and unique to each manufacturer having their own manufacturing processes.

Furthermore, the ethylene oxide sterilization process requires heat and humidity that can increase the migration of substances from the primary packaging to the surface of the device.

Accordingly, in order to adress all risks related to the manufacturing process, primary packaging and sterilization process, all chemical and biological tests are performed on a final finished device, having followed all manufacturing steps, packaged in its primary packaging and sterilization twice by ethylene oxide.

5.2.3 Conclusion on the need of chemical characterisation test

The **Celsite® Implantofix Implant** should be subjected to chemical analysis following the appropriate manufacturing and sterilization processes, encompassing a variety of chemical characterization tests to evaluate the amount, identity, and characteristics of the extractables using methods prescribed in ISO 10993-18: Chemical Characterization of Materials.

As such, the following programs of testing is recommended for the PSU acces port:

- Exhaustive extractions in polar, semi-polar, and non-polar vehicles for determination of total Non-Volatile Residue (NVR).

  These exhaustive extractions should be conducted according to ISO 10993-12. As mentioned in ISO 10993-18: "*The goal of the extraction is to produce an extractables profile that equals or exceeds the leachables generated in clinical use but does not cause deleterious effects to the materials (such as degradation) or the extractables profile (such as chemical alteration of the extractables).*" Exhaustive extractions should be conducted to challenge the materials and extract all chemicals that will leach during the clinical use of the device, but they should not degrade the device or lead to extreme over-exaggeration of these chemicals.

- Fourier Transform Infrared (FTIR) spectroscopy of residues obtained from the test article extracts.
- Determination of trace metals/elements using Inductively Coupled Plasma - Mass Spectrometry / Optical Emission Spectroscopy (ICP-MS) (polar extract only).
- Gas Chromatography with Mass Spectrometry (GC-MS) for detection of Semi-Volatile Organic Compounds (SVOCs) from the test article extracts.
- Ultra-Performance Liquid Chromatography with Mass Spectrometry (UPLC-MS) for detection of Non-Volatile Organic Compounds (NVOCs) from the test article extracts.
- Headspace GC-MS for detection of Volatile Organic Compounds (VOCs).

Once obtained, the chemical characterization data should be subjected to toxicological evaluation per ISO 10993-17 and ISO/TS 21726 to determine if there are unmitigated risks associated with patient exposure to extractable chemicals during clinical use. These data serve a number of useful purposes in the toxicological evaluation of medical devices including:

- The chemical characterization data obtained will allow for extractable chemicals to be identified, in some cases quantified, and toxicologically evaluated. These data will enable a more thorough evaluation to be performed with less leveraged data.
- Chemical characterization will provide the baseline data set useful for supporting the equivalence assessments with other devices.
- Chemical characterization and toxicological profile as obtained can be used for future device iterations for purposes of conducting equivalence assessments, and further considered not to repeat biological testing conducted.

5.3 Chemical characterization test

As different combinations of access ports/catheter/screw connector exist in the Celsite® Implantofix Access port Systems portfolio, one combination of Access Port, Catheter and screw connector has been tested. The other catheters, or representative catheters, have been tested separately.

TABLE 13: SUMMARY OF TEST CONFIGURATIONS

Component	Reference	Designation	Tested through
Access port	Every references		C0221M : Celsite® Implantofix® access port with GR3 catheter
Screw connector	Every references		
Catheter	A3601	Silicone catheter M	C0226L : Catheter DL (A3636) + Connection ring (A3657)
	A3524	PUR catheter 5F/37	C0227L : Catheter C (A3521)

5.3.1 Celsite® Implantofix access port with GR3 catheter (C0221M)

Summary of tests performed

In 2021, Chemical characterization has been conducted on a Celsite® Implantofix access port with GR3 catheter and screw connector. The chemical characterization study was performed to identify and quantitate the extractables and/or leachables that may be released from the test article. This study was conducted based on guidance provided in ISO 10993-18, Biological evaluation of medical devices - Part 18: Chemical characterization of medical device materials within a risk management process.

The chemical characterization report and protocol, are registered under Test Report Number C0221M[36].

A brief summary of the chemical analysis conducted is provided in Table 14 below:

TABLE 14: SUMMARY OF CHEMICAL CHARACTERIZATION TEST C0221N

Date of tests	October 2020 to September 2021
Test article	Celsite® Implantofix access port with GR3 catheter
Extraction	Exhaustive extraction using PW EtOH and 50%EtOH Repeated extraction cycles at 50°C for 72h until exhaustion is reached
ISO standard	ISO 10993-18 (2020) ISO 10993-12 (2021)
Chemical analysis conducted	The resulting extracts were analyzed by FTIR spectroscopy, GC-MS and UPLC-UV-MS. The PW test article extract was additionally analyzed by HS-GC-MS and ICP-MS/OES and IC.
ISO 17025 accreditation or equivalence of the testing laboratory	**Name and Address of the laboratory:** NAMSA - 6750 Wales Road - Northwood - OH 43619 – USA **Certificate number (name of certifying body / validity date)** Certificate n°AT-2561 v005 (ANSI National Accreditation Board / 03.09.2022)[37]

Justification of the selection of the test article

The test article is representative of all Implantofix Access ports and screw connectors as they are made with the same raw materials and with the same manufacturing process.

The test article was packaged in its primary packaging (PET thermoformed tray and Tyvek lid).

As the packed finished device can be re-sterilized if necessary, biocompatibility tests are conducted on test articles sterilized twice by ethylene oxide.

Relevance of the tests performed

The chemical characterization tests were conducted in line with the most recent ISO 10993-18 standard.

Since the conduction of the tests, the following change was implemented:

- **A1007:** Following its production shutdown by the supplier, the green colorant (1% of the polyurethane) used in the screw connector is removed. No other colorant is added, and the polyurethane will stay translucent. This modification has no impact on the biological evaluation of the Celsite® Implantofix Implant as it consists in the removal of an additive, and therefore the removal of the associated toxicological risks. The test article used in the chemical characterization and biological tests stays representative of the Celsite® Implantofix Implant as it is a worst-case.

Chemical/Toxicological Risk Assessment

Based on the data obtained from the chemical characterization testing, a Chemical Risk Assessment is conducted in order to determine if there are extractable substances at levels that could present a toxicological risk.

The Chemical Risk Assessment (CRA), including the description of the methodology for calculating the safety margins, is registered under number CRA 021[38]. The CVs of the authors and reviewers are annexed to the CRA.

Overall, these chemical characterization results obtained on the **Celsite® Implantofix access port with GR3 catheter** indicate that the likelihood of an adverse patient reaction from exposure to each leachable chemical is minimal and associated toxicological risks are acceptable. The test article being representative of the **Celsite® Implantofix Implant**, it can be concluded that the toxicological risks associated with the **Celsite® Implantofix Implant** are acceptable as well.

The acceptability of some systemic toxicity concerns can be further confirmed by biological testing or historical data.

5.3.2 Catheter DL A3636 + Connection ring DL A3657 (C0226)

Summary of Tests performed

In 2021, Chemical characterization has been conducted on a Catheter DL A3636 + Connection ring DL A3657. The chemical characterization study was performed to identify and quantitate the extractables and/or leachables that may be released from the test article. This study was conducted based on guidance provided in ISO 10993-18, Biological evaluation of medical devices - Part 18: Chemical characterization of medical device materials within a risk management process.

The chemical characterization report and protocol, are registered under Test Report Number C0226L[39].

A brief summary of the chemical analysis conducted is provided in Table 15 below:

TABLE 15: SUMMARY OF CHEMICAL CHARACTERIZATION TEST C0226L

Date of tests	April to September 2021
Test article	Catheter DL A3636 + Connection ring DL A3657
Extraction	Exhaustive extraction using PW and 50%EtOH Repeated extraction cycles at 50°C for 72h until exhaustion is reached
ISO standard	ISO 10993-18 (2020) ISO 10993-12 (2021)
Chemical analysis conducted	The resulting extracts were analyzed by FTIR spectroscopy, GC-MS and UPLC-UV-MS. The PW test article extract was additionally analyzed by HS-GC-MS and ICP-MS/OES and IC.
ISO 17025 accreditation or equivalence of the testing laboratory	**Name and Address of the laboratory:** NAMSA - 6750 Wales Road - Northwood - OH 43619 – USA **Certificate number (name of certifying body / validity date)** Certificate n°AT-2561 v005 (ANSI National Accreditation Board / 03.09.2022)[37]

Justification of the selection of the test article

The test article is representative of Catheter M A3601 as they are made with the same raw materials and with the same manufacturing process.

The test article was packaged in its primary packaging (PET thermoformed tray and Tyvek lid).

As the packed finished device can be re-sterilized if necessary, biocompatibility tests are conducted on test articles sterilized twice by ethylene oxide.

Relevance of the tests performed

The chemical characterization tests were conducted in line with the most recent ISO 10993-18 standard.

Chemical/Toxicological Risk Assessment

The chemical characterization tests were conducted in line with the most recent ISO 10993-18 standard.

Based on the data obtained from the chemical characterization testing, a Chemical Risk Assessment is conducted in order to determine if there are extractable substances at levels that could present a toxicological risk.

The Chemical Risk Assessment (CRA), including the description of the methodology for calculating the safety margins, is registered under number CRA028[40]. The CVs of the authors and reviewers are annexed to the CRA.

Overall, these chemical characterization results obtained on the **Catheter DL + Connection ring** indicate that the likelihood of an adverse patient reaction from exposure to each leachable chemical is minimal and associated toxicological risks are acceptable. The test article being representative of the **Celsite® Implantofix Implant**, it can be concluded that the toxicological risks associated with the **Celsite® Implantofix Implant** are acceptable as well.
The acceptability of some systemic toxicity concerns can be further confirmed by biological testing or historical data.

5.3.3 Catheter C A3521 (C0227L)

Summary of Tests performed

In 2021, Chemical characterization has been conducted on a Catheter C (A3521). The chemical characterization study was performed to identify and quantitate the extractables and/or leachables that may be released from the test article. This study was conducted based on guidance provided in ISO 10993-18, Biological evaluation of medical devices - Part 18: Chemical characterization of medical device materials within a risk management process. The chemical characterization report and protocol, are register under Test Report Number C0227L[41].

A brief summary of the chemical analysis conducted is provided below:

TABLE 16: SUMMARY OF CHEMICAL CHARACTERIZATION TEST C0227L

Test article	Catheter C (A3521)
Extraction	Exhaustive extraction using PW, 50%EtOH and Hex Repeated extraction cycles at 50°C for 72h until exhaustivity was reached
ISO standard	ISO 10993-18 (2020) ISO 10993-12 (2021)
Chemical analysis conducted	The resulting extracts were analyzed by FTIR spectroscopy, GC-MS and UPLC-UV-MS. The PW test article extract was additionally analyzed by HS-GC-MS and ICP-MS/OES and IC.
Date of tests	April to September 2021

Justification of the selection of the test article

The test article is representative of Catheter 5F/37 (A3524) as they are made with the same raw materials and with the same manufacturing process.

The test article was packaged in its primary packaging (PET thermoformed tray and Tyvek lid).

As the packed finished device can be re-sterilized if necessary, biocompatibility tests are conducted on test articles sterilized twice by ethylene oxide.

Relevance of the tests performed

The chemical characterization tests were conducted in line with the most recent ISO 10993-18 standard.

Chemical/Toxicological Risk Assessment

Based on the data obtained from the chemical characterization testing, a Chemical Risk Assessment is conducted in order to determine if there are extractable substances at levels that could present a toxicological risk.

The Chemical Risk Assessment, including the description of the methodology for calculating the safety margins, is registered under number CRA 026[42]. The CVs of the authors and reviewers are annexed to the CRA.

Overall, these chemical characterization results obtained on the **Catheter C** indicate that the likelihood of an adverse patient reaction from exposure to each leachable chemical is minimal and associated toxicological risks are acceptable. The test article being representative of the **Celsite® Implantofix Implant**, it can be concluded that the toxicological risks associated with the **Celsite® Implantofix Implant** are acceptable as well.

The acceptability of some systemic toxicity concerns can be further confirmed by biological testing or historical data.

5.4 Potential of degradation

Conducting a systematic evaluation of the potential and observed degradation of this medical device is not considered to be relevant through the design and performance of in vitro degradation studies for the following reasons:

- As stipulated in ISO 10993-1 subclause 4.2, in the selection of materials to be used in the medical device manufacture, the first consideration shall be fitness for purpose with regard to characteristics and properties of the material. Accordingly, the materials selected to manufacture components intended to be in long-term contact with tissues and biological fluids were appropriately selected for their intended use;

- As stipulated in ISO 10993-9 preliminary considerations, it is neither necessary nor practical to conduct degradation studies for all medical devices. However, it shall include a review of the literature and/or documented clinical experience.
    - The raw materials used for long term implantable parts are not innovative.
    - No new polymeric materials, or modified polymeric materials with unknown degradation behavior in body contact, are expected that would be requested to be screened per ISO 10993-13.
    - These long-term implanted raw materials are well-known and widely used in medical devices, including access ports and catheters.
    - No chemical degradation of PSU after implantation in the body has been identified in the analyzed toxicological data during the literature analysis. The unalloyed Titanium is provided compliant to its standards.
- Silicone remain a widely used raw materials for septums included in access ports.
- The potential degradation related to the components in silicon is already identified. Corresponding standards for evaluating the degradation products of the polymeric components, including silicone, is ISO 10993-13. Per ISO 10993-13 scope, only degradation products generated by a chemical alteration of the finished polymeric device is to be considered. A comprehensive chemical characterization is conducted. No biological hazards from potential degradation products are expected that would not be identified through the exhaustive extractions conditions with extractions in solvents of different polarities. Additionally, degradation inducted by mechanical stress, wear or electromagnetic radiation or biological factors is to be evaluated according to the principles of ISO 10993-1; ISO 10993-16 and ISO 10993-17.
- Degradation products are well-known and clinically accepted. Considering a demonstrated history of acceptable clinical experience, actual available data is considered as sufficient for no further testing pertaining to degradation purposes is needed.

6. Biological testing program

6.1 Justification on the need or not to perform biological evaluation tests

The **Celsite® Implantofix Implant** is a medical device in long-term contact (>30 days) with circulating blood and tissues. Based on ISO 10993-1 and considering the 2020 FDA Guidance, the biological risks that need to be addressed are cytotoxicity, sensitization, irritation or intracutaneous reactivity, material mediated pyrogenicity, acute systemic toxicity, hemolysis, thrombosis, subacute/subchronic toxicity, chronic toxicity, implantation effects, genotoxicity, carcinogenicity, as well as immunotoxicity.

From a regulatory standpoint, while all of the endpoints specified above must be considered on the **Celsite® Implantofix Implant**, they can be addressed in a number of different ways, including testing, written justification or risk assessment if relevant information is already available.

For the raw materials of the **Celsite® Implantofix Implant** that contact the patient for a long-term duration, the expected risks are limited. However, based on the performed analysis, some hazards were identified: concerns for irritation and allergic reactions are reported. Furthermore, degradation products from the polyurethane or silicone need to be evaluated. As such, additional information to evaluate risk (for example, determination of exposure) are needed.

In the manufacturing process, for processing aids for which information was available, some raw materials are associated with health concerns such as irritation. The chemical characterization with toxicological risk assessment should be conducted to address any residuals that may remain on the device.

While few concerns are expected related to the primary packaging raw materials, sterilization is conducted with ethylene oxide, which is known to be associated with highly reactive residues. Furthermore, the ethylene oxide sterilization process requires heat and humidity that can increase the migration of substances from the primary packaging to the surface of the device. In order to adress these risks, all chemical and biological tests are conducted on a finished device packaged in the primary packaging and sterilized twice with ethylene oxide.

In regards to irritation and sensitization concerns, this endpoint can be mitigated by biological testing. Assuming favorable results are obtained, the data would support that exposure (and, thus, risk) is low, if any.

Concerns associated with reproductive/developmental toxicities and suspicion of damaging fertility are associated with hazards that are *a priori* not relevant to be addressed for the **Celsite® Implantofix Implant**. However, due to the long-term duration of contact, these risks need to be evaluated according to the level of patient exposure. Therefore, detections of these substances have to be addressed in a quantitative manner. The chemical characterization with associated toxicological risk analysis was conducted for addressing these risks.

Comprehensive chemical characterization of test articles representative of the **Celsite® Implantofix Implant** were conducted. Evaluation covered total NVRs with FTIR of the residues after exhaustive extractions in three different vehicles, ICP (MS or OES), SVOCs, NVOCs and VOCs detections. Such testing allow to establish the toxicological profile of the **Celsite® Implantofix Implant**.

Overall, these chemical characterization results obtained indicate that the likelihood of an adverse patient reaction from exposure to each leachable chemical is minimal and associated toxicological risks are acceptable.The acceptability of some systemic toxicity concerns can be further confirmed by biological testing or historical data.

Finally, no information is available on the toxicological risks of the combination of substances. While the risk of having an additive interaction between the substances is low, the acceptability of the toxicological risks can be confirmed by biological testing.

Considering all of the above, the proposed risk-control strategy to address the biocompatibility of the **Celsite® Implantofix Implant** is detailed in the following section.

6.2 Determination of the testing program

Considering that the biological testing is intended to support the safety of these components, tests shall be carried out according to Good Laboratory Practice (GLP) principles (*per* ISO 10993-1, subclause 6.3.1 and the FDA Biocompatibility Guidance, section IV, B, 4

- **Cytotoxicity** (ISO 10993-5): Highly sensitive *in vitro* test intended to screen biologically harmful extractables in the absence of protective mechanisms that normally assist cells within the body. A quantitative method such as MTT or MTS is recommended to be conducted. An extraction at 37°C

for 72 hours (instead of 24 hours *per* ISO 10993-5, subclause 4.2.3.2.) is recommended for devices with prolonged or long-term contact duration, as a worst-case.

- **Sensitization** (ISO 10993-10): *In vivo* test intended to determine whether significant quantities of leachables/extractables could induce delayed-type hypersensitivity (Type IV) reactions. The ISO 10993-10 recommends a Guinea Pig Maximization test to performed as it is the most sentitive method and the occlusive system test is recommended for topical products. Two extractions in polar and apolar extraction vehicles (such as 0,9% sodium chloride and sesame oil) must be performed. An extraction at 50°C for 72 hours is recommended as *per* ISO 10993-12.
- **Intracutaneous reactivity** (ISO 10993-10: 2010, ISO 10993-23: 2021): *In vivo* test intended to evaluate irritation potential of leachables/extractables after a short-term exposure by the intracutaneous route. For medical devices used as implants, the ISO 10993-23, paragraph 7.3 recommends *in vivo* intracutaneous injection tests which are more relevant. An extraction at 50°C for 72 hours using polar and apolar extraction vehicles (such as 0,9% sodium chloride and sesame oil) is recommended as *per* ISO 10993-12.
- **Acute systemic toxicity** (ISO 10993-11): *In vivo* test intended to estimate, during a period of less than 24 hours, the potential harmful effects of systemic exposure to leachables/extractables. An extraction at 50°C for 72 hours using polar and apolar extraction vehicles (such as 0,9% sodium chloride and sesame oil) is recommended as *per* ISO 10993-12.
- **Material-mediated pyrogenicity** (ISO 10993-11 and Regional Pharmacopoeia): *in vivo* test intended to detect substances inducing a febrile reaction in the patient. Rabbit Pyrogenicity tests following the European and US Pharmacopeia are recommended.
- **Hemocompatibiliy:**
    - **Hemolysis** (ISO 10993-4 and ASTM F756): This *in vitro* test is intended to evaluate the potential of the test article to cause red blood cell hemolysis. A hemolytic index is determined. This test is recommended to be carried out with both direct contact method and indirect contact method on devices in final finished forms.
    - **Complement activation assay – SC5b-9** (ISO 10993-4): *In vitro* test intended to evaluate the potential of leachables/extractables to activate the complement system, which consists of small blood proteins circulating as inactive precursors in blood and intended to facilitate the destruction and removal of foreign materials by the immune system. SC5b-9: unique to the terminal pathway of the complement cascade.
    - **Partial thromboplastin time** (ISO 10993-4): The partial thromboplastin time is the clotting time of recalcified citrated plasma upon the addition of partial thromboplastin which does not contain an activator. Partial thromboplastin is a phospholipid suspension usually extracted from tissue thromboplastin, the homogenate from mammalian brain or lung. Shortening of the PTT following contact with a material under standard conditions indicates activation of the intrinsic coagulation pathway of blood coagulation. PTT testing compared to a legally marketed reference device (e.g., a device with history of prolonged use in similar procedures) is advised to provide further assurance of the device thrombogenic potential.

- **Genotoxicity** (ISO 10993-3): Tests intended to determine gene mutations, changes in chromosome structure and number, and other DNA or gene toxicities that could cause leachables/extractables in extracts. As *per* ISO 10993-3, two *in vitro* tests have to be considered:
    - **Ames test** (Bacterial Reverse Mutation Assay, OECD 471): Highly sensitive *in vitro* test intended to detect gene mutations affecting a small portion of the DNA molecule including frameshifts and base-pair substitutions following contact with test-extracts. Extraction in two vehicles of different polarities is recommended.
    - **Mouse lymphoma assay** (MLA, OECD 490): *In vitro* test intended to evaluate whether extracts of the test-article induce chromosomal damage in the mouse lymphoma L5178Y/Tk+/- cell line, heterozygous at the thymidine kinase (tk) locus. Extraction in two vehicles of different polarities is recommended.

6.3 Rationale for tests not completed

Clause 6.2: "*ISO 10993-2 applies to any in vivo testing being considered. Additional in vivo testing shall not be carried out where:*

1) *Results are available from relevant studies that have been carried out previously; or*
2) *The existing non-clinical and clinical data, including history of safe use, meet the requirements of biological evaluation and therefore further animal testing would be unethical. In assessing the relevance of data on prior use of a material to the biological evaluation, the level of confidence in the historical data should be taken into account.*

Rationale for the omission of tests is provided below:

Subacute/subchronic/chronic systemic toxicity (ISO 10993-11): *In vivo* tests intended to evaluate the adverse effects on blood and/or organs/tissues resulting from absorption, distribution and metabolism of potential toxic leachables/extractables due to continuous exposure for short- (subacute), mid-(subchronic) or long-term (chronic). These tests are not considered as mandatory to be conducted based on the following:

- The materials are well-known and during the literature search that was conducted as part of this risk assessment, there were no studies or reports found indicating that these materials would pose a significant risk of systemic toxicity in animals or humans.
- The chemical characterization and toxicological evaluation results allow to adequately address the subacute/subchronic/chronic systemic toxicity risk.
- The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.

Implantation (ISO 10993-6): *In vivo* tests intended to evaluate the local tissue responses (irritation, inflammation, etc.) following clinically-relevant exposure to the device after different durations of exposure. This biological risk is not considered as mandatory to evaluate through additional testing based on the following:

- The materials are well-known and during the literature search that was conducted as part of this risk assessment, there were no studies or reports found indicating that these materials would pose a significant risk of local intolerance in animals or humans.

- The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.

Carcinogenicity (ISO 10993-3): *In vivo* test intended to determine the tumorigenic potential of a test article. Carcinogenicity test should only be performed if there is a reasonable suspicion that carcinogenicity of a device is a true risk. *Per* ISO 10993-1 subclause 6.2.2.1 "*If there is no information from other sources, testing the potential carcinogenicity of the material/device shall be considered. However, it is rare for carcinogenicity tests to be considered appropriate for medical devices (see ISO 10993-3)*." *Per* ISO 10993-3 subclause 6.1 "*The decision to perform a* (carcinogenicity) *test shall be justified on the basis of an assessment of the risk of carcinogenesis arising from the use of the medical device. Carcinogenicity testing shall not be performed when risks can be adequately assessed or managed without generating new carcinogenicity test data*." This biological risk is not considered as mandatory to evaluate through additional animal testing based on the following:

- The materials are well known materials with clinical use history in medical devices similar to this device
- The chemical testing and toxicological assessments provided results allowing to adequately address the carcinogenicity risk
- The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.

Immunotoxicity (ISO 10993-20): Immunotoxicity is defined by as the study of the adverse health effects that result, directly or indirectly, from the interaction of xenobiotics with the immune system. Potential immunological outcomes for implanted devices include hypersensitivity, chronic inflammation, immunosuppression, immunostimulation, and autoimmunity. ISO/TS 10993-20 notes that the likelihood of immunogenicity is greatest with foreign proteins. An evaluation of immunotoxicity in line with the guidance given in ISO/TS 10993-20 is indicated only where circumstances suggestive of possible immunotoxicity apply. Immunotoxicity testing is unwarranted for the following reasons:

- The materials are not known to have adverse immunological effects.
- The chemical testing and toxicological assessments provided results allowing to adequately address the immunotoxicity risk.
- The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.

Reproductive/developmental toxicity (ISO 10993-3): This *in vivo* test is intended to evaluate the potential effects of medical devices, materials and/or their extracts on reproductive function, embryonic development (teratogenicity), and prenatal and early postnatal development. Clause 7.1 of this standard mentions that "*The decision to perform a test shall be justified on the basis of an assessment of the risk of reproductive and developmental toxicity arising from the use of the medical device*". Moreover, clause 6.2.2.12 of ISO 10993-1 specifies that "*Reproductive /developmental toxicity tests or bio-assays shall only be conducted when the device has potential impact on the reproductive potential of the subject* […]". This biological risk is not considered as mandatory to evaluate through animal testing based on the following:

- the materials are well-known and during the literature search that was conducted as part of this risk assessment, there were no studies or reports found indicating that these materials would pose a significant risk of reproductive/developmental toxicity in animals or humans.

- the chemical testing programs and toxicological assessments provided results showing the absence of extractables/leachables known to cause reproductive/developmental toxicity.
- The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.

Degradation (ISO 10993-9, ISO 10993-13, ISO 10993-15, ISO 10993-16)

The rationale for not completing *in vitro* degradation studies is provided in Section 5.4.

6.4 Summary of testing program

The testing program is summarized in Table 17.

TABLE 17: METHODS OF CONSIDERATION OF BIOLOGICAL ENDPOINTS FOR THE CELSITE® IMPLANTOFIX IMPLANT

Biological Testing program*		Celsite® Implantofix Implant (contact > 30 days)
Cytotoxicity		T
Sensitization		T
Intracutaneous reactivity		T
Systemic Toxicity	Material Mediated Pyrogenicity	T
	Acute Systemic Toxicity	T
	Subacute/subchronic (13 weeks) Toxicity	R
	Chronic Toxicity	R
Hemocompatibility	Hemolysis (direct and indirect)	T
	Partial thromboplastin time	T
	SC5b-9 Complement Activation Assay	T
Local Tissue Effects (Implantation)		R
Genotoxicity	Bacterial Reverse Mutation Assay	T
	Mouse Lymphoma Assay	T
Carcinogenicity		R
Immunotoxicity		R
Reproductive/developmental toxicity		R
Degradation		R

(T): Test recommended to be performed. Biological tests should be conducted according to GLP regulations (FDA GLP).
(R): Rationalized test; this test can be omitted based on assumptions or consideration (see Section 6.3)
(*): *Per* ISO 10993-12, it is recommended that extractions be performed at 50°C for 72 hours (apart for cytotoxicity).

PART 2 : BIOLOGICAL EVALUATION REPORT

6.5 Biological tests performed

As different combinations of access ports/catheter/screw connector exist in the Celsite® Implantofix Access port Systems portfolio, one combination of Access Port, Catheter and screw connector has been tested. The other catheters, or representative catheters, have been tested separately.

TABLE 18: SUMMARY OF TEST CONFIGURATIONS

Component	Reference	Designation	Tested through
Access port	Every references		C0221: Celsite® Implantofix® access port with GR3 catheter
Screw connector	Every references		
Catheter	A3601	Silicone catheter M	C0226: Catheter DL (A3636) + Connection ring (A3657)
	A3524	PUR catheter 5F/37	C0227: Catheter C (A3521)

6.5.1 Celsite® Implantofix Access Port With Gr3 Catheter (C0221)

Biocompatibility tests have been conducted on a Concept access port A3062 with L catheter A3600 and Connection ring A3632. A high-level summary of all tests conducted is presented in the following Table 19.

TABLE 19: BIOLOGICAL TESTS CONDUCTED ON CONCEPT ACCESS PORT A3062 WITH L CATHETER A3600 AND CONNECTION RING A3632

Study Number	Standard Applied	Biocompatibility Test	Extraction conditions	Results
Test Article: Celsite® Implantofix® access port with GR3 catheter, (Access port I3130, Screw connector: I2025, Catheter: I3060) 2xEO sterile				
C0221A[43]	ISO 10993-4 (2017) ASTM F756-17 (2017) ISO 10993-12 (2012)	ISO 10993 and ASTM Hemolysis	CMF-DPBS, 50°C, 72h, 3cm²/mL	Pass
C0221B[44]	ISO 10993-10 (2010) ISO 10993-12 (2012)	ISO Intracutaneous Study in Rabbits (Two Extracts)	SC, SO, 50°C, 72h, 3cm²/mL	Pass
C0221C[45]	ISO 10993-5 (2012) ISO 10993-12 (2012)	ISO MTS cytotoxicity test	EMEM10, 37°C, 72h, 3cm²/mL	Pass
C0221D[46]	ISO 10993-11 (2017) ISO 10993-12 (2012)	ISO Acute Systemic Toxicity Study in Mice (Two Extracts)	SC, SO, 50°C, 72h, 3cm²/mL	Pass
C0221E[47]	ISO 10993-11 (2017) ISO 10993-12 (2012) EP 10th, 2.6.8 (2020)	Rabbit Pyrogen Study according to European Pharmacopeia	SC, 50°C, 72h, 3cm²/mL	Pass
C0221F[48]	ISO 10993-11 (2017) ISO 10993-12 (2012) USP 42 NF 37 (2019)	Rabbit Pyrogen Study according to United States Pharmacopeia 42 - National Formulary 37 (USP)	SC, 50°C, 72h, 3cm²/mL	Pass
C0221G[49]	ISO 10993-4 (2017) ISO 10993-12 (2012)	SC5b-9 Complement Activation Assay	NHS, 37°C, 1h, 3cm²/mL	Pass
C0221H[50]	ISO 10993-3 (2014) OECD 471 (1997) ISO 10993-12 (2012)	Genotoxicity: Bacterial Reverse Mutation Study (Two extracts)	SC, PEG, 72h, 50°C, 3cm²/mL	Pass
C0221I[51]	ISO 10993-10 (2010) ISO 10993-12 (2012)	ISO Guinea Pig Maximization Sensitization Test (Two Extracts)	SC, SO, 50°C, 72h, 3cm²/mL	Pass
C0221J[52]	ISO 10993-3 (2014) ISO 10993-12 (2012)	Genotoxicity: Mouse Lymphoma Assay (Two Extracts)	RPMI, 37°C, 72h, 3cm²/mL PEG, 50°C, 72h, 3cm²/mL	Pass

Study Number	Standard Applied	Biocompatibility Test	Extraction conditions	Results
C0221K[53]	ASTM F2382	ASTM Partial Thromboplastin Time	Citrated human plasma, 37°C, 15min	Pass

EMEM10: Eagle Minimum Essential Medium with 10% fetal bovine serum; CMF-DPBS: Calcium-Magnesium free Dulbecco's phosphate-buffered saline; SC: Sodium Chloride 0,9%; SO: Sesame oil; NHS: Normal Human Serum; PEG: Polyethylene glycol; RPMI: Roswell Park Memorial Institute medium

Justification of the selection of the test article

The biological tests were conducted on the same test article than the chemical characterization test.

The justification of the selection of the test article is presented in Section 5.3.1

Relevance of the tests performed

Irritation has been conducted per ISO 10993-10 (2010) requirements while now ISO 10993-23 (2021) is applicable. ISO 10993-23 (2021) is presenting alternative *in vitro* testing to *in vivo* ones for evaluation irritation. However, when implants are considered, *in vivo* tests remain recommended.

Sensitization has been conducted per ISO 10993-10:2010 while now ISO 10993-10 (2021) is applicable. ISO 10993-10 (2021) has been updated to reflect the split between the irritation and the sensitization guidance. Therefore, the creation of the ISO 10993-23 and the update of the ISO 10993-10 are not impacting the validity of the results.

Regarding extraction methods, ISO 10993-12:2021 currently applies. Considering that the updates to ISO 10993-12 mainly discuss the conditions for extractions of devices that are intended to degrade, no impacts of the update of this standard are seen on the tests conducted.

6.5.2 Catheter DL A3636 + Connection ring DL A3657 (C0226)

Biocompatibility tests have been conducted on a Catheter DL A3636 + Connection ring DL A3657. A high-level summary of all tests conducted is presented in the following Table 20.

TABLE 20: BIOLOGICAL TESTS CONDUCTED ON CATHETER DL A3636 + CONNECTION RING DL A3657

Study Number	Standard Applied	Biocompatibility Test	Extraction conditions	Results
Test article: Catheter DL A3636 + Connection ring DL A3657 (2xEO sterile)				
C0226A[54]	ISO 10993-11 (2017) ISO 10993-12 (2012)	ISO Acute Systemic Toxicity Study in Mice (Two Extracts)	SC, SO, 50°C, 72h, 6cm²/mL	Pass
C0226B[55]	ISO 10993-11 (2017) ISO 10993-12 (2012) EP 10th, 2.6.8 (2020)	Rabbit Pyrogen Study according to European Pharmacopeia	SC, 50°C, 72h, 6cm²/mL	Pass
C0226C[56]	ISO 10993-11 (2017) ISO 10993-12 (2012) USP 42 NF 37 (2019)	Rabbit Pyrogen Study according to United States Pharmacopeia 42 - National Formulary 37 (USP)	SC, 50°C, 72h, 6cm²/mL	Pass
C0226D[57]	ASTM F2382	ASTM Partial Thromboplastin Time	Citrated human plasma, 37°C, 15min	Pass
C0226E[58]	ISO 10993-10 (2010) ISO 10993-12 (2012)	ISO Intracutaneous Study in Rabbits (Two Extracts)	SC, SO, 50°C, 72h, 6cm²/mL	Pass

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Study Number	Standard Applied	Biocompatibility Test	Extraction conditions	Results
C0226F[59]	ISO 10993-5 (2012) ISO 10993-12 (2012)	ISO MTS cytotoxicity test	EMEM10, 37°C, 72h, 6cm²/mL	Pass
C0226G[60]	ISO 10993-4 (2017) ASTM F756-17 (2017) ISO 10993-12 (2012)	ISO 10993 and ASTM Hemolysis	CMF-DPBS, 50°C, 72h, 6cm²/mL	Pass
C0226H[61]	ISO 10993-3 (2014) OECD 471 (1997) ISO 10993-12 (2012)	Genotoxicity: Bacterial Reverse Mutation Study (Two extracts)	SC, PEG, 72h, 50°C, 6cm²/mL	Pass
C0226I[62]	ISO 10993-4 (2017) ISO 10993-12 (2012)	SC5b-9 Complement Activation Assay	NHS, 37°C, 1h, 6cm²/mL	Pass
C0226J[63]	ISO 10993-10 (2010) ISO 10993-12 (2012)	ISO Guinea Pig Maximization Sensitization Test (Two Extracts)	SC, SO, 50°C, 72h, 6cm²/mL	Pass
C0226K[64]	ISO 10993-3 (2014) ISO 10993-12 (2012)	Genotoxicity: Mouse Lymphoma Assay (Two Extracts)	RPMI, 37°C, 72h, 6cm²/mL PEG, 50°C, 72h, 6cm²/mL	Pass

EMEM10: Eagle Minimum Essential Medium with 10% fetal bovine serum; CMF-DPBS: Calcium-Magnesium free Dulbecco's phosphate-buffered saline; SC: Sodium Chloride 0,9%; SO: Sesame oil; NHS: Normal Human Serum; PEG: Polyethylene glycol; RPMI: Roswell Park Memorial Institute medium

Justification of the selection of the test article

The biological tests were conducted on the same test article than the chemical characterization test.

The justification of the selection of the test article is presented in Section 5.3.2

Relevance of the tests performed

Irritation has been conducted per ISO 10993-10 (2010) requirements while now ISO 10993-23 (2021) is applicable. ISO 10993-23 (2021) is presenting alternative *in vitro* testing to *in vivo* ones for evaluation irritation. However, when implants are considered, *in vivo* tests remain recommended.

Sensitization has been conducted per ISO 10993-10:2010 while now ISO 10993-10 (2021) is applicable. ISO 10993-10 (2021) has been updated to reflect the split between the irritation and the sensitization guidance. Therefore, the creation of the ISO 10993-23 and the update of the ISO 10993-10 are not impacting the validity of the results.

Regarding extraction methods, ISO 10993-12:2021 currently applies. Considering that the updates to ISO 10993-12 mainly discuss the conditions for extractions of devices that are intended to degrade, no impacts of the update of this standard are seen on the tests conducted.

6.5.3 Catheter C A3521 (C0227)

Biocompatibility tests have been conducted on a Catheter C A3521. A high-level summary of all tests conducted is presented in the following Table 21.

TABLE 21: BIOLOGICAL TESTS CONDUCTED ON CATHETER C A3521

Study Number	Standard Applied	Biocompatibility Test	Extraction conditions	Results
Test article: Catheter C (A3521) (2xEO sterile)				
C0227A[65]	ISO 10993-4 (2017) ASTM F756 (2017) ISO 10993-12 (2012)	ISO 10993 and ASTM Hemolysis	CMF-DPBS, 50°C, 72h, 6cm²/mL	Pass
C0279B[66]	ISO 10993-4 (2017) ASTM F2382 (2018)	ASTM Partial Thromboplastin Time	citrated human plasma, 37°C, 15min, 6cm²/mL	Pass
C0227C[67]	ISO 10993-11 (2017) ISO 10993-12 (2012) ASTM F750 (2012)	ISO Acute Systemic Toxicity Study in Mice (Two Extracts)	SC, SO, 50°C, 72h, 6cm²/mL	Pass
C0227D[68]	ISO 10993-5 (2009) ISO 10993-12 (2012)	ISO MTS cytotoxicity test	EMEM, 37°C, 72h, 6cm²/mL	Pass
C0227E[69]	ISO 10993-10 (2010) ISO 10993-12 (2012)	ISO Intracutaneous Study in Rabbits (Two Extracts)	SC, SO, 50°C, 72h, 6cm²/mL	Pass
C0227F[70]	ISO 10993-11 (2017) ISO 10993-12 (2012) EP 10th, 2.6.8 (2020)	Rabbit Pyrogen Study according to European Pharmacopeia	SC, 50°C, 72h, 6cm²/mL	Pass
C0227G[71]	ISO 10993-11 (2017) ISO 10993-12 (2012) USP 43 NF 38 (2020)	Rabbit Pyrogen Study according to United States Pharmacopeia 43 - National Formulary 38 (USP)	SC, 50°C, 72h, 6cm²/mL	Pass
C0227H[72]	ISO 10993-3 (2014) ISO 10993-12 (2012)	Genotoxicity: Bacterial Reverse Mutation Study (Two extracts)	SC, DMSO, 50°C, 72h, 6cm²/mL	Pass
C0227I[73]	ISO 10993-4 (2017) ISO 10993-12 (2012)	SC5b-9 Complement Activation Assay	NHS, 37°C, 1h, 6cm²/mL	Pass
C0227J[74]	ISO 10993-2 (2006) ISO 10993-10 (2010) ISO 10993-12 (2012)	ISO Guinea Pig Maximization Sensitization Test (Two Extracts)	SC, SO, 50°C, 72h, 6cm²/mL	Pass
C0227K[75]	ISO 10993-3 (2014) ISO 10993-12 (2012)	Genotoxicity: Mouse Lymphoma Assay (Two Extracts)	RPMI, 37°C, 72h, 6cm²/mL PEG, 50°C, 72h, 6cm²/mL	Pass

EMEM10: Eagle Minimum Essential Medium with 10% fetal bovine serum; CMF-DPBS: Calcium-Magnesium free Dulbecco's phosphate-buffered saline; SC: Sodium Chloride 0,9%; SO: Sesame oil; NHS: Normal Human Serum; PEG: Polyethylene glycol; RPMI: Roswell Park Memorial Institute medium

Justification of the selection of the test article

The biological tests were conducted on the same test article than the chemical characterization test.

The justification of the selection of the test article is presented in Section 5.3.4

Relevance of the tests performed

Irritation has been conducted per ISO 10993-10 (2010) requirements while now ISO 10993-23 (2021) is applicable. ISO 10993-23 (2021) is presenting alternative *in vitro* testing to *in vivo* ones for evaluation irritation. However, when implants are considered, *in vivo* tests remain recommended.

Sensitization has been conducted per ISO 10993-10:2010 while now ISO 10993-10 (2021) is applicable. ISO 10993-10 (2021) has been updated to reflect the split between the irritation and the sensitization guidance.

Therefore, the creation of the ISO 10993-23 and the update of the ISO 10993-10 are not impacting the validity of the results.

Regarding extraction methods, ISO 10993-12:2021 currently applies. Considering that the updates to ISO 10993-12 mainly discuss the conditions for extractions of devices that are intended to degrade, no impacts of the update of this standard are seen on the tests conducted.

Evidence of the ISO 17025 accreditation or equivalence of the testing laboratory:

All biocompatibility tests have been performed by laboratories with accreditation to ISO 17025. The list of laboratories and certificate identification number are presented in the following Table 22.

TABLE 22: LABORATORIES ACCREDITATION

Name and Address of the laboratory	SMQ certificate identification number (name of certifying body / validity date)
NAMSA 115 Chemin de L'Islon 38670 Chasse-sur-Rhône - France	Certificate n°1-0628 rev8[76] (Cofrac / 31.12.2022)

7. Overall analysis of the results

7.1 Summary and analysis of all the data described and/or results obtained in the previous stages

This biological risk assessment is supported by information on the **Celsite® Implantofix Implant** materials of construction, available information on the manufacturing process, gathered toxicological data on these materials, and existing data on this device.

Considering all gathered data, this risk assessment indicates that a comprehensive biocompatibility testing program was needed to establish the biocompatibility profile of the **Celsite® Implantofix Implant** including chemical characterization and biological testing.

Biological testing provided information on the risks associated with cytotoxicity, irritation, sensitization, pyrogenicity, hemocompatibility, which are information not provided through chemical characterization and toxicological analysis, as well as information on the risks of mutagenicity / genotoxicity.

Based on the results of the biological testing, the **Celsite® Implantofix Implant** is considered non-cytotoxic, non-sensitizing, non-irritant, non-pyrogenic, not an acute toxicant and not mutagenic. Furthermore, the device is considered to be hemocompatible based on the results of testing (ASTM Hemolysis Study, SC5b-9 Complement Activation and ASTM Partial Thromboplastin Time tests).

Finally, the data obtained from the toxicological risk assessment conducted on the substances identified by chemical characterization should allow to consider that the risks of carcinogenicity, Subacute/subchronic/chronic systemic toxicity and immunotoxicity are acceptable.

TABLE 23: SUMMARY OF RISK ASSESSMENT

Endpoint	Toxicological Risk	Risk control
Cytotoxicity	Acceptable	Cytotoxicity tests have been performed on representative products. Test articles has shown no signs of cell toxicity at L-929 mouse fibroblasts. The final products meet the requirements of the test and are judged no cytotoxic
Sensitization	Acceptable	Sensitization tests have been performed on a representative products. The topical application of the 0,9% sodium chloride and sesame oil extracts of the test articles did not induce deayed sensitization in the guinea pig. The test articles are not considered a sensitizer in the guinea pig maximization model.
Intracutaneous reactivity	Acceptable	Intracutaneous reactivity tests have been performed on a representative product. The test article met the requirement of the test following intracutaneous injection in rabbits after extraction in 0,9% sodium chloride and sesame oil. The test article is not considered as irritant.
Pyrogenicity	Acceptable	Pyrogenicity tests have been performed on representative products. The test articles met the requirement of the USP and EP. The test articles were judged non-pyrogenic.
Acute Systemic Toxicity	Acceptable	Acute systemic toxicity tests have been performed on a representative product. Under the conditions of this study, there was no mortality or evidence of systemic toxicity from the 0,9% sodium chloride and sesame oil extracts injected into mice. Each test article extract met the requirements of the study.
Subacute/subchronic/ChronicToxicity	Acceptable	The materials used are well-known and there were no studies or reports found in the literature indicating that these materials would pose a significant risk of systemic toxicity in animals or humans. The chemical characterization conducted on a representative device and the toxicological evaluation results allow to adequately address the subacute/subchronic/chronic systemic toxicity risk. The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing
Hemocompatibility (Hemolysis)	Acceptable	Hemolysis tests (indirect and indirect) have been performed on representative products. Under the conditions of this study, the mean hemolytic index for the test article extract was of 0.1 %. The test articles extracts were non hemolytic. Partial Prothromboplastin time assays and SC5b-9 Complement activation assays have been performed on representative products. The test articles were not considered to be a potential activator of the complement system and it met the requirements of the PTT tests.
Local Tissue Effects (Implantation)	Acceptable	The materials used are well-known and there were no studies or reports found in the literature indicating that these materials would pose a significant risk of systemic toxicity in animals or humans. The chemical characterization conducted on a representative device and the toxicological evaluation results allow to demonstrate that no susbtances known to have local effects were detected at levels that could cause toxicological risks. The requirements of ISO 10993-1 and ISO 10993-2 that advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.
Genotoxicity	Acceptable	Genotoxicity tests were performed on representative products. In the Bacterial Reverse Mutation Study, the 0,9% sodium chloride and polyethylene glycol test article extracts were considered to be non-mutagenic to *S. typhimurium* tester strains TA98, TAl00, TA1535, and TA1537, and to *E. coli WP2uvrA* tester strain. In the Mouse Lymphoma Assay, The RPMI and PEG test article extracts did not cause any increase in the mean mutant frequency of the L5178Y/TK" cell line greater than the mutant frequency of the control blank + the Global Evaluation Factor (126), in the presence or absence of metabolic activation. The test articles are considered non mutagenic.
Carcinogenicity	Acceptable	The materials used are well-known and there were no studies or reports found in the literature indicating that these materials would pose a significant risk of systemic toxicity in animals or humans. The chemical characterization conducted on a representative device and the toxicological evaluation results allow to adequately address the carcinogenicity risk.

Endpoint	Toxicological Risk	Risk control
		The requirements of ISO 10993-1 and ISO 10993-2 advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing
Immunotoxicity	Acceptable	The materials used are not known to have adverse immunological effects. The chemical characterization conducted on a representative device and the toxicological evaluation results allow to adequately address the immunotoxicity risk. The requirements of ISO 10993-1 and ISO 10993-2 advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.
Reproductive/ developmental toxicity	Acceptable	The materials used are well-known and there were no studies or reports found in the literature indicating that these materials would pose a significant risk of systemic toxicity in animals or humans. The chemical characterization conducted on a representative device and the toxicological evaluation results allow to adequately address the reproductive/developmental toxicity risk. The requirements of ISO 10993-1 and ISO 10993-2 advocate to consider every alternative method to animal use whenever possible and avoid unnecessary animal testing.
Degradation	Acceptable	The materials used are well-known and widely used in implantable medical devices. The degradation products from the polymeric components are well known and the chemical characterization The chemical characterization conducted on a representative device and the toxicological evaluation results allow to adequately address the toxicological risks related to these potential degradation products.

7.2 Reassessment of Risk

This risk assessment is valid for the current iteration of the **Celsite® Implantofix Implant**. It applies to devices manufactured using the current processes and techniques.

As specified in the Clause 4.9 of the current ISO 10993-1, the biological risk assessment of the **Celsite® Implantofix Implant** shall be re-evaluated if any of the following occur:

- any change in the source or in the specification of the materials used in the manufacture of the product,
- any change in the formulation, processing, primary packaging, or sterilization of the product,
- any change in the manufacturer's instructions or expectations concerning storage, e.g. changes in shelf life and/or transport,
- any change in the intended use of the product,
- any evidence that the product can produce adverse biological effects when used in humans.

During the redaction of the document, the following change was implemented:

- **A1005**: 3 references of Celsite® Implantofix are discontinued since May 26th, 2024, and therefore removed from the scope of this Biological Evaluation report: configurations ST SET SIL 6F IV (ref 4438704), ST PUR 5F IV (ref 4430263) and ST PUR 5F IA (ref 4438817) and associated catheters A3523 and I3060 are removed of the report.

The following changes were implemented and described in the previous editions of this Biological Evaluation Report:

- **A706**: the Celsite® IMPLANTOFIX references 04438620 and 04438663 were deleted as the Splitocan needle 8433925 contained in the kits will no longer be produced.
- **A792**: change of Pellethane 2363-80AE and 2363-90A manufacturing site
- The polyurethane Pellethane 2363-90A and Pellethane 2363-80AE, provided by Lubrizol LifeSciences, are used in the manufacture of the Celsite® Implantofix access ports. This change is the consequence of the supplier notification indicating that the production of Pellethane® 2363 Thermoplastic Polyurethane (TPU) medical polymer products will be transferred from the current manufacturing site in La Porte, TX to the Lubrizol production site in Avon Lake, OH. According to the data provided by Lubrizol, physical, analytical and performance testing were conducted in order to demonstrate the equivalence in physical, chemical and processing properties of the Avon Lake produced material to the La Porte product77. Biocompatibility tests have been conducted by Lubrizol on the two materials manufactured in both fabrication sites[78]. Overall, the tests performed by Lubrizol demonstrate that the change of manufacturing sites has no impact on the biocompatibility of Pellethane® 2363-90A and 2363-80AE.
- **X505**: In 2020, DuPont announce a new brand name for their silicone-based healthcare procudt range. Dow brand names including DOW CORNING™ and Silastic™ become Liveo™. Documents including the new brand name (MSDS, TDS, PRI) are available. The raw material composition table was updated to include the rebranding.
  Furthermore, analytical characterization has been performed by DuPont to better define the levels of Octamethylcyclotetrasiloxane (D4), Decamethylcyclopentasiloxane (D5) and dodecamethylcyclohexasiloxane (D6) present in several Liveo™ products. In particular, the levels of cyclic siloxanes present in elastomers have been measured lower than currently documented in their compositions. Therefore, the compositions, safety data sheets and the labels of these products were adapted to better reflect the real levels of these substances in the Liveo™ products. The formulations, compositions and manufacturing processes of these products have not changed.
  This modification has no impact on the current biological risk assessment because there is no change in the formulation of manufacturing of the product.
- **A912**: The green ink printing on the lids is replaced by a laser printed label attached to the breather bag and called “internal label”. To ensure traceability during sterilization, a small label with the batch number and the product article code number will be applied on the breather bag. No leachables from the small label are expected to migrate inside the breather bag, through the tyvek lid and onto the device surface. Therefore, this change has no impact on the biological safety of the finished device.
- **A953**: The condition of post-curing of the septum has been changed from 150°C for 2 hours to 120°C for 1 hour 30 minutes. While the decrease of curing temperature may lead to the increase of volatile substances components still in the septum, it is unlikely to have a significant impact on the potential leachables for the following reasons:
  - The supplier information indicate that the principal volatile components are short-chained polydimethylsiloxanes fluids.
  - As the shortest linear siloxane (hexamethyldisiloxane) is volatile with a boiling point of 100°C, a change of post-curing temperature from 150°C to 120°C is not expected to have significant impact on the volatile residues in the septum.
  - As shown in the Toxicological Risk Assessment conducted, during chemical characterization of the access port containing the septum, the potential exposure to siloxanes was greatly exaggerated by the extraction condition. Overall, the toxicological risks related to the siloxanes are still acceptable.

- Biocompatibility tests have been conducted on Silastic® Q7-4750 Biomedical grade ETR elastomer (former name of Liveo™ Q7-4750 Biomedical Grade ETR Elastomer) without post-curing[79], overall, the test article met the requirements of the ISO 10993 standards.
- The post-curing process is primarily intended to ensure that the material is fully cured and free from any residual curing agents or unreacted monomers that may potentially migrate out of the material over time. As long as the material is fully cured and free from any residual curing agents or unreacted monomers, a lower post-curing temperature of 120°C would not significantly impact the material's biocompatibility.

- **A957Bis**: The manufacturer Crozet SAS informed us of the change of a processing aid used in the manufacture of the exit cannula, housing rings and cups. The polishing agent WHEELABRATOR GROUP B11 was replaced by RODASTEL 30. The impact of this change was evaluated through test report 23-001[80]. After chemical analysis, no residues of RODASTEL 30 were identified above the detection liimit after extraction of a worst-case test article. It can be concluded that no biological risk related to residues of RODASTEL 30 are expected, this modification has no impact on the biological safety of the finished device.
- **A976**: The name of the luer connector was change from TECHNYL A216 NATURAL FA to TECHNYL SAFE A216FC NC. The composition of the raw material is unchanged, only the name of the material. This modification as no impact on the biological safety of the finished device.
- **A1007**: Following its production shutdown by the supplier, the green colorant (1% of the polyurethane) used in the screw connector is removed. No other colorant is added, and the polyurethane will stay translucent. This modification has no impact on the biological evaluation of the Celsite® Implantofix Implant as it consists in the removal of an additive, and therefore the removal of the associated toxicological risks. The test article used in the chemical characterization and biological tests stays representative of the Celsite® Implantofix Implant as it is a worst-case.

8. References

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Referenced documents:

1 **3_DK:** Safety Data Sheet of EVIVA® EV-26000

2 **3_DK(TDS):** Technical Data Sheet of Eviva® EV-26000

3 **210101_FR24_DL_EVIVA**: Declaration letter on Regulation (EU) 2017/745 for Eviva® PSU products, by Solvay. Written January 1, 2021.

4 **3_OS:** Material Safety Data Sheet of Liveo™ Q7-4750 Biomedical Grade ETR Elastomer

5 **3_OS(TDS):** Technical Data Sheet of Liveo™ Q7-4750 Biomedical Grade ETR Elastomer

6 **220602_FR24_COMP_X1450**: Declaration Letter on composition of Liveo™ Q7-4750 Biomedical Grade ETR Elastomer

7 **201030_FR24_PRI_X1450**: Product Regulatory Information of Liveo™ Q7-4750 Biomedical Grade ETR Elastomer

8 **3_FD**: material Certificate Sortie Implantofix Gr.3

9 **3_AG**: Safety Data Sheet of PELLETHANE® 2363-80AE NAT

10 **3_IG**: Material Safety Data Sheet of Liveo™ RX65R Medical Grade Tubing Radiopaque

11 **3_IG(TDS)**: Technical Data Sheet of Liveo™ RX65R Medical Grade Tubing Radiopaque

12 **210119_FR24_PRI_A1601**: Product Regulatory Information of Liveo™ RX65R Medical Grade Tubing Radiopaque

13 **220602_FR24_PRI_A1601**: Declaration letter on composition of Liveo™ RX65R Medical Grade Tubing Radiopaque

14 **3_DW**: Safety Data Sheet of MED-6608-2

15 **3_EZ:** Material Safety Data Sheet of TECHNYL SAFE A 216FC NC

16 **3_EZ(TDS):** Technical Data Sheet of TECHNYL SAFE A 216FC NC

17 **3_IB**: Material Safety Data Sheet of Tecothane® TT2095A-B20

18 **3_DL_3_EL_3_IB(TDS):** Technical Data Sheet of Tecothane® TPU

19 **PRO-03-2022:** Literature Search and Review plan for Polysulfone

20 **REP-03-2022**: Final Report, Literature review for Polysulfone

21 **PRO-19-2022**: Literature Search and Review plan for Titanium Dioxide

22 **REP-19-2022**: Final Report, Literature review for Titanium Dioxide

23 **PRO-07-2022**: Literature Search and Review plan for Zinc Oxide

24 **REP-07-2022**: Final Report, Literature review for Zinc Oxide

25 **PRO-06-2022**: Literature Search and Review plan for Silicone

26 **REP-06-2022**: Final Report, Literature review for Silicone

27 **PRO-11-2022**: Literature Search and Review plan for Stainless Steel

28 **REP-11-2023**: Final Report, Literature review for Stainless Steel

29 **PRO-17-2022**: Literature Search and Review plan for Barium Sulfate

30 **REP-17-2022**: Literature Search and Review plan for Barium Sulfate

31 **PRO-21-2022**: Literature Search and Review plan for Polyurethane

32 **REP-21-2022**: Literature review for Polyurethane

33 **PA0002_MSD_Ethanol 70_ZCP211_Alcogroup**: Material Safety Data Sheet of ALCOOL ETHYLIQUE 70%

34 **PA0021_MSD_neodisher MediClean forte_ZCP488**: Material Safety Data Sheet of NEODISHER MEDICLEAN FORTE

35 **PA0033_MSD_TOPKLEAN MC 20D**: Material Safety Data Sheet of SOLVANT A3 TOPKLEAN MC20D

36 **C0221M_CHEM** : Test Report: ISO 10993-18 - Chemical Characterization - Celsite® Implantofix access port with GR3 catheter

37 **GLP_AT-2561_v005_NAMSA Northwood**: ISO 17025 certificate N° AT-2561 Version 005 of NAMSA Northwood

38 **CRA021**: Chemical Risk Assessment related to C0221M

39 **C0226L_CHEM** : Test Report: ISO 10993-18 - Chemical Characterization - Catheter DL + Connection ring

40 **CRA028**: Chemical Risk Assessment related to C0226L

41 **C0227L_CHEM**: Test Report - ISO 10993-18 - Chemical Characterization Program – Catheter C (A3521)

42 **CRA026** : Chemical Risk Assessment related C0227L

43 **C0221A_HEM_Hemolysis**: Test Report - ISO 10993 and ASTM Hemolysis - Celsite® Implantofix® access port with GR3 catheter

44 **C0221B_IRR_IC**: Test Report - ISO Intracutaneous Study in Rabbits (Two Extracts) - Celsite® Implantofix® access port with GR3 catheter

45 **C0221C_CYT_MTS**: Test Report - ISO MTS cytotoxicity - Celsite® Implantofix® access port with GR3 catheter

46 **C0221D_TOX_Acute**: Test Report - ISO Acute Systemic Toxicity Study in Mice (Two Extracts) - Celsite® Implantofix® access port with GR3 catheter

47 **C0221E_PYR_EP**: Test Report - Rabbit Pyrogen Study according to European Pharmacopeia - Celsite® Implantofix® access port with GR3 catheter

48 **C0221F_PYR_USP**: Test Report - Rabbit Pyrogen Study according to United States Pharmacopeia 42 - National Formulary 37 (USP) - Celsite® Implantofix® access port with GR3 catheter

49 **C0221G_HEM_CAA**: Test Report - SC5b-9 Complement Activation Assay - Celsite® Implantofix® access port with GR3 catheter

50 **C0221H_GEN_Ames**: Test Report - Genotoxicity: Bacterial Reverse Mutation Study (Two extracts) - Celsite® Implantofix® access port with GR3 catheter

51 **C0221I_SEN_GPM**: Test Report - ISO Guinea Pig Maximization Sensitization Test (Two Extracts) - Celsite® Implantofix® access port with GR3 catheter

52 **C0221J_GEN_MLA**: Test Report - Genotoxicity: Mouse Lymphoma Assay (Two Extracts) - Celsite® Implantofix® access port with GR3 catheter

53 **C0221K_HEM_PTT**: Test Report - ASTM Partial Thromboplastin Time - Celsite® Implantofix® access port with GR3 catheter

54 **C0226A_TOX_Acute**: Test Report - ISO Acute Systemic Toxicity Study in Mice (Two Extracts) - Catheter DL + Connection ring

55 **C0226B_PYR_EP**: Test Report - Rabbit Pyrogen Study according to European Pharmacopeia- Catheter DL + Connection ring

56 **C0226C_PYR_USP**: Test Report - Rabbit Pyrogen Study according to United States Pharmacopeia 42 - National Formulary 37 (USP) - Catheter DL + Connection ring

57 **C0226D_HEM_PTT**: Test Report - ASTM Partial Thromboplastin Time - Catheter DL + Connection ring

58 **C0226E_IRR_IC**: Test Report - ISO Intracutaneous Study in Rabbits (Two Extracts) - Catheter DL + Connection ring

59 **C0226F_CYT_MTS**: Test Report - ISO MTS cytotoxicity - Catheter DL + Connection ring

60 **C0226G_HEM_Hemo**: Test Report - Rabbit Pyrogen Study according to European Pharmacopeia - Catheter DL + Connection ring

61 **C0226H_GEN_Ames**: Test Report - Genotoxicity: Bacterial Reverse Mutation Study (Two extracts) - Catheter DL + Connection ring

62 **C0226I_HEM_CAA**: Test Report - SC5b-9 Complement Activation Assay - Catheter DL + Connection ring

63 **C0226J_SEN_GPM**: Test Report - ISO Guinea Pig Maximization Sensitization Test (Two Extracts) - Catheter DL + Connection ring

[64] **C0226J_GEN_MLA**: Test Report - Genotoxicity: Mouse Lymphoma Assay (Two Extracts) - Catheter DL + Connection ring

65 **C0227A_HEM_Hemolysis**: Test Report - ISO 10993 and ASTM Hemolysis - Catheter C (A3521)

66 **C0227B_HEM_PTT**: Test Report - ASTM Partial Thromboplastin Time - Catheter C (A3521)

67 **C0227C_TOX_Acute**: Test Report - ISO Acute Systemic Toxicity Study in Mice (Two Extracts) - Catheter C (A3521)

68 **C0227D_CYT_MTS**: Test Report - ISO MTS cytotoxicity – Catheter C (A3521)

69 **C0227E_IRR_IC**: Test Report - ISO Intracutaneous Study in Rabbits (Two Extracts) - Catheter C (A3521)

70 **C0227F_PYR_EP**: Test Report - Rabbit Pyrogen Study according to European Pharmacopeia - Catheter C (A3521)

71 **C0227G_PYR_USP**: Test Report - Rabbit Pyrogen Study according to United States Pharmacopeia 42 - National Formulary 37 (USP) - Catheter C (A3521)

72 **C0227H_GEN_Ames**: Test Report - Genotoxicity: Bacterial Reverse Mutation Study (Two extracts) - Catheter C (A3521)

73 **C0227I_HEM_CAA**: Test Report - SC5b-9 Complement Activation Assay - Catheter C (A3521)

74 **C0227J_SEN_GPM**: Test Report - ISO Guinea Pig Maximization Sensitization Test (Two Extracts) - Catheter C (A3521)

75 **C0227K_GEN_MLA** : Test Report - Genotoxicity: Mouse Lymphoma Assay (Two Extracts) - Catheter C (A3521)

76 **GLP_1-0628_rev8_NAMSA Lyon**: ISO 17025 certificate N° 1-0628 rev.8 of NAMSA Chasse-sur-Rhône

77 Lubrizol Data Package

78 Lubrizol Biocompatibility tests

79 **Q7-4750 Health Data**: Summary of Health Data for Silastic® Q7-4750 Biomedical grade ETR elastomer

80 **CR23-001**: Analyse d'impact du remplacement de l'adjuvant WHEELABRATOR GROUP B11 par RODASTEL 30